

LATENTSPRING: HARMONIC SOURCES AND PHYSICAL CORRECTIONS FOR MOLECULAR GENERATION

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ABSTRACT

Generating a molecule from atomic composition requires determining both its connectivity and three-dimensional coordinates. In this paper, we propose LatentSpring, which improves molecular generation through two designs. A normalized harmonic-mixture source uses random auxiliary trees and atomic length scales to organize the starting coordinates without known chemical bonds. A small equivariant network learns physical corrections to a frozen flow matching (FM) model’s predicted endpoints from force feedback. On 64 unseen compositions, LatentSpring reaches 24.00% joint yield, the fraction of outputs that are graph-valid and have GFN2-xTB force RMS at most 5 eV/Å, compared with 8.75% for GAGA and 8.15% for EDM. A component study using one network evaluation per velocity prediction shows that the harmonic source adds 4.35 percentage points to joint yield when combined with physical correction. The correction also improves the flow matching model in each of three new training repetitions. Expressing its action in coordinate space allows a physical correction head trained using FM-generated trajectory states to act directly on diffusion noise predictions: transfer to GAGA improves joint yield by 16.54 points across those three repetitions, without retraining the correction. Generation uses neural evaluations, with no energy calls or quantum geometry optimization.

1 INTRODUCTION

An atomic composition tells us which atoms a molecule contains, but not how they are connected or arranged in three spatial dimensions. In this paper, we investigate the task of molecular generation from this input, together with the molecule’s total charge and spin multiplicity. The model returns atomic coordinates. We then infer bonds from those coordinates to evaluate the generated structure. Unlike conformation generation with a known molecular graph, this task leaves both connectivity and geometry to be determined.

Flow matching learns to transform samples from a source distribution into a target distribution (Lipman et al., 2023). An isotropic Gaussian is a simple choice, but it places all atoms on a common spatial scale without using atom-specific geometric information. Harmonic sources instead correlate atom positions through virtual springs. Existing graph-conditioned methods can construct these springs from known chemical bonds. The question here is how to obtain a useful structured source when those bonds are not given.

In this paper, we propose LatentSpring, which uses a random tree to organize the starting coordinates. We sample Gaussian displacements along its edges, with scales determined by the connected atoms’ radii, and obtain atom positions by adding these displacements along the tree. A new tree is drawn for every sample. Its edges are not proposed chemical bonds: in the flow matching model, the tree is used only for initialization, and the flow can generate a structure with different connectivity, including rings. Averaging over the trees gives a normalized mixture of harmonic networks. We combine this source with an equivariant coordinate network and geometry self-conditioning: the network makes a provisional geometry prediction, then uses it to refine its velocity prediction.

Learning physical geometry poses a second challenge when the chemical graph is unknown. An atomic potential can provide forces directly from coordinates, but evaluating it at every sampling step

adds repeated cost. We learn a small correction from these forces offline. The frozen parent predicts a provisional final geometry, and an energy model supplies forces at that geometry during training. A bounded equivariant head learns the additional velocity needed to shift the prediction along those forces. At generation time, only the neural networks are evaluated. Expressing the head’s action as a change to the provisional endpoint also allows it to correct a diffusion model’s noise prediction.

In summary, our contributions are threefold:

- We construct a normalized harmonic-mixture source from atomic composition. Random auxiliary trees introduce atom-dependent geometric correlations while leaving the generated chemical connectivity to the learned flow.
- We learn a bounded equivariant correction from forces at predicted endpoints. The coordinate-space interface connects this correction to flow velocities and diffusion noise predictions, with the backbone fixed during correction training and neural-network evaluations during sampling.
- We evaluate generation quality, the contribution of each design, and reuse of learned correction weights across generative objectives. Three new training pairs confirm the correction’s benefit, and the FM-trained heads improve GAGA joint yield by 16.54 percentage points without target-specific head training.

2 RELATED WORK

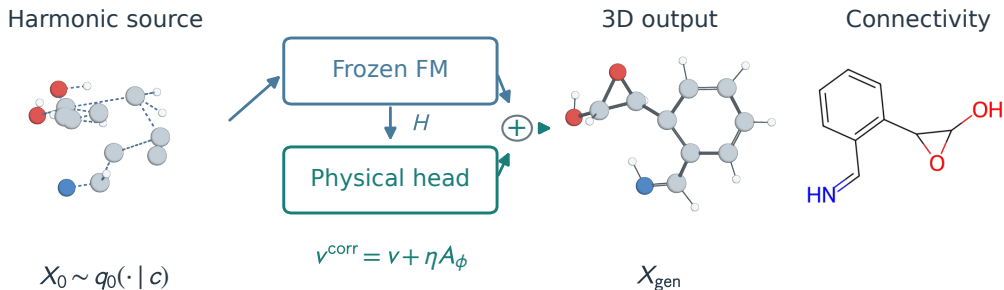
Molecular flows and source distributions. ConfGF and GeoDiff generate conformations for a supplied chemical graph (Shi et al., 2021; Xu et al., 2022). Our main comparisons use the EGNN underlying EDM (Hoogetboom et al., 2022); additional source experiments use FlowMol3’s coordinate network and self-conditioning modules (Dunn & Koes, 2026). Equivariance means that rotating the input rotates the predicted vectors, while relabeling identical atoms relabels their outputs. HarmonicFlow combines harmonic sources with self-conditioned flow matching for ligand docking; FlowSite extends this approach to binding-site design (Stark et al., 2024). ET-Flow also builds a harmonic source from a supplied molecular graph (Hassan et al., 2024). These methods know the chemical connectivity at the input. Our source instead averages over auxiliary spatial trees and requires only atomic composition. Other approaches to source adaptation include guidance with a frozen vector field (Wang et al., 2026b) and learned condition-dependent sources (Kim et al., 2026).

Distributions over trees. Factored tree distributions allow tractable probabilities and learning over trees (Meila & Jaakkola, 2000; Duan & Dunson, 2023). Gaussian mixtures defined through random spanning-tree differences have also been used for imaging inverse problems (Everink, 2026). We use isotropic edge displacements and atomic length scales to define a distribution over centered molecular coordinates. Our source experiments compare this mixture with isotropic Gaussian, shell-tree, and covariance-matched Gaussian alternatives.

Physical feedback and nonequilibrium work. OMol25 provides molecular structures, energies, and forces across electronic conditions (Levine et al., 2025). Energy-weighted flow matching (Dern et al., 2025), importance-sampling-based training (Midgley et al., 2023), and physical-reward fine-tuning of molecular diffusion models (Zhou et al., 2025) provide ways to incorporate physical information into generation. Simulation data also support protein fine-tuning (Lu et al., 2024), and physical proxies can guide residual correction of frozen video flow fields (Wang et al., 2026a). FLEXDOCK regularizes predicted docking endpoints with a geometry-based potential (Corso et al., 2025). ConfDiff learns a force-guidance network for protein conformations (Wang et al., 2024), while Lai et al. (2026) apply MMFF94 guidance during molecular flow and diffusion sampling. We learn bounded equivariant corrections from bond-free forces at generated provisional endpoints, keeping the parent fixed and using neural evaluations during generation. A separate extension uses nonequilibrium work identities to reweight local physical targets (Jarzynski, 1997; Hummer & Szabo, 2001; Vaikuntanathan & Jarzynski, 2008; Jarzynski, 2002). Related learned escorting supports multistate free-energy estimation (Holdijk et al., 2026). We test local distribution recovery with controlled molecular rotors and subsequent neural training.

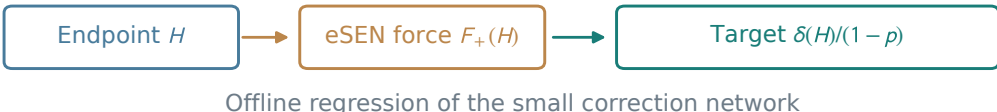
A Composition to raw coordinates

$C_9H_9NO_2$ · neutral singlet



Dashed links: auxiliary springs

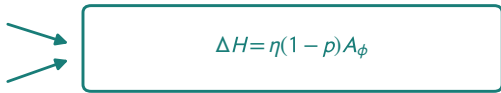
B Learn the correction from forces



C Apply the same coordinate correction

FM: $H = X + (1-p)v$

Diffusion: $H = (X - \sigma_t \epsilon)/\alpha_t$



7,106 parameters · Sampling energy queries: 0

Figure 1: LatentSpring generates coordinates from atomic composition. (A) A sampled source and a raw $C_9H_9NO_2$ output share camera and scale. Dashed links are auxiliary springs; the 2D inset shows inferred connectivity. The example is selected from the validation illustration panel in Appendix C. (B) Endpoint forces provide offline regression targets. (C) Each sampler defines its own endpoint, and the head specifies a coordinate displacement. The head receives current coordinates, endpoint coordinates, atom identities and progress.

3 A HARMONIC SOURCE WITH LATENT CONNECTIVITY

We represent the input as $c = ((z_i)_{i=1}^N, Q, S)$, where z_i is the atomic number of each atom, Q is the total charge in elementary-charge units, and S is the spin multiplicity. Repeated elements appear once per atom. A neutral singlet has $Q = 0$ and $S = 1$. For example, C_2H_6O specifies two carbon atoms, six hydrogens, and one oxygen, but allows different bond connectivities. Bonds are inferred from coordinates for evaluation, not supplied as training labels.

The output coordinates form $X \in \mathbb{R}^{N \times 3}$, where row x_i is atom i ’s position in angstroms. We subtract the mean atom position to remove translation. The centered coordinates lie in $\mathcal{H}_N = \{X : \sum_i x_i = 0\}$, a space with $3(N-1)$ independent coordinates.

Sampling a tree. We first sample a spanning tree T over the atoms: a connected graph with $N-1$ edges and no cycles. This tree tells us how to place the starting coordinates; it does not specify the final chemical bonds. Each possible edge (i, j) has a positive weight, or affinity, $a_{ij} = a_{ji}$. We sample a tree with probability proportional to the product of its edge weights:

$$p(T | c) = \frac{\prod_{(i,j) \in T} a_{ij}}{\tau(a)}, \quad \tau(a) = \sum_T \prod_{(i,j) \in T} a_{ij}. \quad (1)$$

The sum $\tau(a)$ normalizes this distribution over all spanning trees. We use $a_{ij} = u_i u_j$, where $u_i = 0.05$ for H and F/Cl/Br/I, and $u_i = \max(1, \text{default valence}(z_i) - 1)$ otherwise. Larger u_i favors more tree edges incident to atom i . Small weights for hydrogen and halogens favor terminal positions. Default valence is an element-level bonding capacity, counting bond order; these weights require no molecule-specific bonds.

Placing atoms along the tree. For each tree edge, we independently sample a three-dimensional Gaussian displacement. Its scale depends on ℓ_{ij} , the sum of the two atoms’ covalent radii, which estimates a typical single-bond separation. We set $\sigma_{ij}^2 = \ell_{ij}^2 \exp(s^2)/3$ with $s = 0.2$. The edge displacement $d \in \mathbb{R}^3$ has density

$$h_{ij}(d) = (2\pi\sigma_{ij}^2)^{-3/2} \exp\left[-\frac{\|d\|^2}{2\sigma_{ij}^2}\right]. \quad (2)$$

Here σ_{ij} is the standard deviation along each axis. All directions are equally likely, and the radius-based scale determines the typical separation. This choice also matches the edge covariance of the lognormal shell control. The FlowMol implementation samples trees with weighted Wilson walks; the shared EGNN uses weighted Prüfer sequences for the same factorized tree law. We add edge vectors along tree paths and subtract the mean position. We draw a new tree for every sample. The resulting law is unchanged by rotations or relabelings of identical atoms.

The source density. For FM, the tree is unobserved after initialization. Summing over it gives the source density $q_0(X | c)$, a mixture of harmonic networks:

Proposition 1 (Normalized source density). *For normalized isotropic edge densities,*

$$q_0(X | c) = N^{3/2} \frac{\tau(a_{ij} h_{ij}(x_i - x_j))}{\tau(a)} \quad (3)$$

integrates to one over the centered coordinate space $\mathcal{H}_N$.

The numerator sums tree weights after multiplying each edge affinity by its displacement density. The factor $N^{3/2}$ converts independent edge vectors into centered atom coordinates. The matrix-tree theorem evaluates each sum through a determinant rather than enumerating trees; Appendix P derives the normalization. Training only needs samples from the tree-and-edge procedure. It does not evaluate this determinant. Physical learning uses a separate local Gaussian neighborhood around each predicted endpoint.

We call the source *harmonic* because each Gaussian edge density corresponds to a quadratic spring energy. These virtual springs organize the initial random coordinates; the learned flow determines the final molecular geometry.

4 INTEGRATION WITH MOLECULAR FLOW MATCHING

Given the input c , the flow matching model transforms $X_0 \sim q_0(\cdot | c)$ into $X_{\text{gen}} \in \mathcal{H}_N$. The velocity network takes the current coordinates X_t , an embedding of flow time t , and c to predict $v_\theta(X_t, t; c) \in \mathbb{R}^{N \times 3}$: one velocity vector per atom. Integrating this field from $t = 0$ to $t = 1$ produces the coordinates. The latent tree is used only to sample X_0 and is not passed to the network as a bond graph. Energy and force evaluations are used during physical training, not during generation.

Training pairs and velocity targets. We draw X_0 from the harmonic source and take a reference structure X_1 from OMol25. We align the structures by matching atoms of the same type and rotating them to reduce their coordinate displacement. We then apply the same random rotation and the same random relabeling of identical atoms to both structures. The rotation is uniform over orientations. This shared augmentation preserves the source marginal (Appendix Q).

We use the linear path $X_t = (1 - t)X_0 + tX_1$, with velocity target $V^* = X_1 - X_0$. Thus training is supervised regression from an intermediate structure to the displacement between its paired endpoints. The variable $t \in [0, 1]$ measures progress along this interpolation, not physical time.

Geometry self-conditioning. The main EGNN predicts one centered velocity vector per atom. Self-conditioning evaluates this network twice. The first prediction $v^{(0)}$ gives a provisional endpoint, $\tilde{X}_1 = X_t + (1 - t)v^{(0)}$. The second pass receives its pairwise distances through an invariant edge channel and predicts $v^{(1)}$. Both passes share weights; the provisional geometry is detached before constructing the second-pass distance features.

We supervise both velocity predictions:

$$\mathcal{L}_{\text{FM}} = \frac{1}{2} \mathbb{E} \left[\frac{\|v^{(0)} - V^*\|_F^2 + \|v^{(1)} - V^*\|_F^2}{3N} \right]. \quad (4)$$

The expectation averages over reference structures, source samples, and training times. The loss averages the two passes and the $3N$ coordinate errors, giving each molecule equal weight. Both evaluations receive supervised gradients, while the distance feedback is held fixed during differentiation. Atom identities remain fixed and no chemical bond labels are supplied. The one-pass component ablation omits the second evaluation. Appendix E describes the earlier FlowMol implementation.

Sampling. The main FM uses 32 midpoint steps. Each step evaluates the current coordinates and time, advances a half-step, then evaluates the half-step coordinates at the midpoint time to obtain the full-step update. Two self-conditioning passes per velocity evaluation give $32 \times 2 \times 2 = 128$ backbone calls per output. The physical correction below acts on each final velocity prediction, adding 64 small-network calls. The generated coordinates are returned without a terminal perturbation, energy-based selection, or geometry optimization.

4.1 LEARNING PHYSICAL CHANGES TO PREDICTED ENDPOINTS

A physical force is defined on a geometry, while a generator predicts a velocity or a noise vector. We connect these quantities through the generator’s *provisional endpoint*: its current estimate of the final coordinates. The backbone remains fixed, and a small network learns how physical feedback should change this estimate (Figure 1).

From forces to a regression target. Let X be the current coordinates and p the fraction of the generation schedule completed. For a flow, p is its interpolation time and the endpoint estimate is $H = X + (1 - p)v$, where v is the parent velocity. We evaluate the centered, mirror-averaged eSEN force $F_+(H)$ during training and form

$$\delta(H) = \frac{\sigma_H^2}{k_B T_{\text{phys}}} F_+(H), \quad T_{\text{phys}} = 300 \text{ K}. \quad (5)$$

Here σ_H is a local coordinate width, distinct from diffusion noise. Under a first-order energy approximation, multiplying a Gaussian centered at H by a Boltzmann factor shifts its mean by exactly $\delta(H)$. We choose $\sigma_H \leq 0.03 \text{ \AA}$ so the molecular displacement norm is at most $0.1 \text{ \AA}$. Appendix T gives the derivation. The local shift supplies the regression target for physical correction.

The correction network $A_\phi(X, H, p; c)$ receives current and predicted final coordinates, atom identities, and progress. Symmetric pair coefficients multiply opposite relative vectors, giving an equivariant correction with zero total velocity. Its per-atom norm is bounded by $2p^2 \text{ \AA}$ per unit progress. We fit A_ϕ by squared error to $\delta(H)/(1 - p)$, uniformly rescaling this target if necessary to satisfy the same bound. All recorded training states are retained, including invalid provisional geometries. The final layer starts at zero, so the initial model reproduces the parent. Only the small correction network is trained; no gradients update the backbone.

The same endpoint change for flow and diffusion. A strength η sets the intended endpoint change to $\eta(1 - p)A_\phi$. For diffusion, the parent predicts noise ϵ and its endpoint estimate is $H = (X - \sigma_t \epsilon)/\alpha_t$, where α_t and σ_t are the fixed schedule’s signal and noise scales. These two parameterizations require

$$v^{\text{corr}} = v + \eta A_\phi, \quad \epsilon^{\text{corr}} = \epsilon - \eta \frac{\alpha_t(1 - p)}{\sigma_t} A_\phi. \quad (6)$$

Substitution into either endpoint estimate gives the same displacement $\eta(1 - p)A_\phi$. In GAGA, $p = 1 - t/t_{\text{max}}$, with diffusion noise time t and truncation $t_{\text{max}} = 0.65$. The head-only transfer retains each sampler’s transitions and observation noise. Progress indexes the generation schedule.

The head architecture, bounds and fitting protocol are given in Appendices M and N. Paired backbone updates and work-weighted local targets are studied separately in Appendices L and S.

Conditional hydrogen readout. An optional flow adjusts hydrogen coordinates while holding relative heavy-atom positions fixed. It learns from perturbed reference hydrogens and is activated by a detached-hydrogen contact test. Four midpoint steps add eight network calls (Appendix N.4).

5 EXPERIMENTS

5.1 EXPERIMENTAL SETUP

We study molecular generation conditioned on atomic composition, total charge, and spin multiplicity. The main comparison uses an EGNN backbone trained on 20,000 OMol25 structures and evaluates 64 unseen neutral organic singlet compositions with 17–40 atoms. Each trained model generates 16 structures per composition. We compare Gaussian-source FM, EDM (Hoogeboom et al., 2022), GAGA (Qu et al., 2026), and LatentSpring. The latter combines the harmonic-mixture source with learned physical correction; the optional hydrogen readout is excluded from the main comparison and component ablation.

We report two measures. *Graph validity* is the fraction of outputs for which bond inference produces a connected, sanitized chemical graph with the requested charge. *Joint yield* additionally requires a GFN2-xTB atomic force RMS of at most 5 eV/Å. Both denominators include every generation attempt. All physical evaluations use the generated coordinates, with no geometry optimization or energy-based filtering. Confidence intervals resample compositions and, where specified, fitted models. Appendix B gives implementation details and Appendix F specifies the structural checks.

5.2 COMPARISON WITH GENERATIVE BASELINES

We compare LatentSpring with Gaussian-source flow matching, EDM, and GAGA. LatentSpring combines the harmonic-mixture source and learned physical correction; the baselines use their own generators without these additions. Figure 2 reports both measures on the same compositions.

LatentSpring has the highest mean graph validity and joint yield in this comparison. Its joint yield is 24.00%, compared with 8.75% for GAGA, 8.15% for EDM, and 6.25% for Gaussian FM. The gains over GAGA and EDM are 15.25 and 15.85 percentage points, respectively. The separation is larger for joint yield than for graph validity: more generated structures satisfy both the chemical-graph checks and the physical force threshold.

A stricter screen of bond lengths, angles, steric distances and planarity raises the fraction satisfying both that screen and the force criterion from 1.64% for the FM parent to 4.43% after correction. Appendix C reports the complete geometric diagnostics.

5.3 ABLATION OF THE SOURCE AND PHYSICAL CORRECTION

We separate the two design choices in a four-setting comparison: a Gaussian or harmonic source, each with or without physical correction. These experiments use EGNN flow models with one network pass per velocity prediction. Within each training run, the same frozen correction head is applied to both source variants. This keeps the learned correction fixed while changing the source.

Physical correction improves joint yield with either source: from 6.25 to 19.92% for Gaussian FM and from 7.23 to 24.27% for harmonic FM. The harmonic source alone gives smaller changes. Its contribution is clearer when combined with physical correction: graph validity rises from 20.65 to 25.59%, and joint yield rises from 19.92 to 24.27%. The 4.35-point joint-yield gain has a 95% interval of [1.56, 7.13] when resampling both training runs and compositions. Both runs favor the combination over either component alone. Appendix B reports the individual runs and the four-setting analysis.

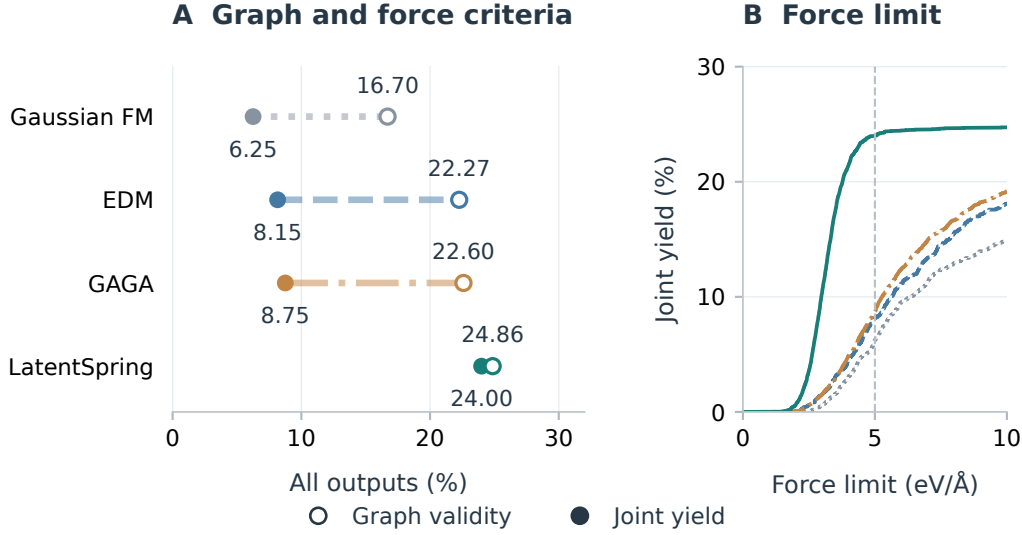


Figure 2: Generation quality on 64 unseen compositions. (A) Open markers show graph validity; filled markers show joint yield at the 5 eV/Å force threshold. The connecting segment counts graph-valid outputs above that threshold. (B) Empirical joint-yield curves computed from every saved force score; the dashed line marks 5 eV/Å. Colors and line styles identify the same methods in both panels. Gaussian FM and EDM use two fits (2,048 outputs per method); GAGA and LatentSpring use five (5,120). The hydrogen readout is disabled.

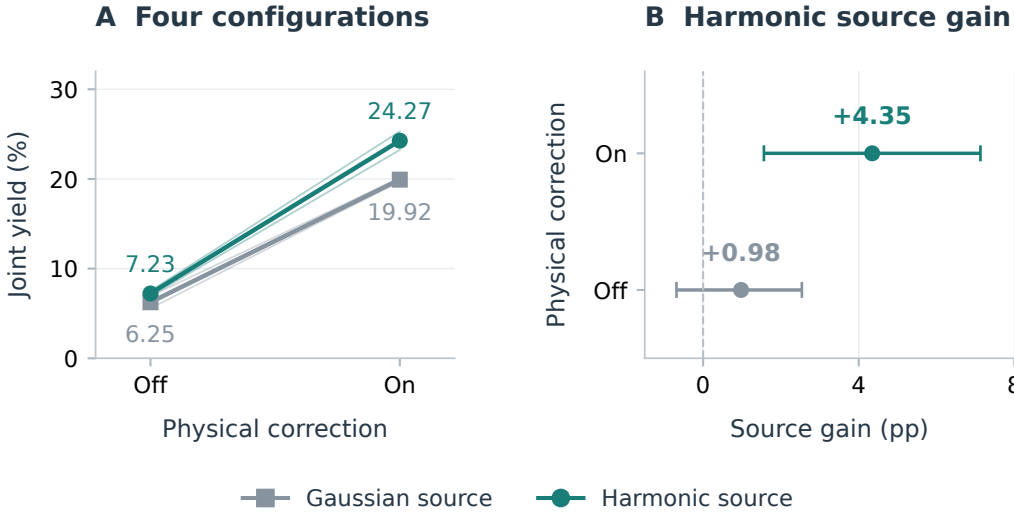


Figure 3: Contributions of the source and physical correction. (A) Mean joint yield for the four configurations; faint lines retain both training fits. (B) Harmonic-minus-Gaussian joint-yield gains, with nominal 95% intervals from resampling fits and compositions. Each configuration uses 2,048 outputs on the same 64 compositions and one network evaluation per velocity prediction.

5.4 TRANSFER ACROSS GENERATIVE MODELS

The endpoint interface allows the learned correction to be reused with another sampler. We exchange the FM and GAGA heads within each trained pair, preserving their tensors and calibration strength. The target retains its backbone, source distribution and sampling schedule. Figure 4 compares each parent with its own head and with the transferred head.

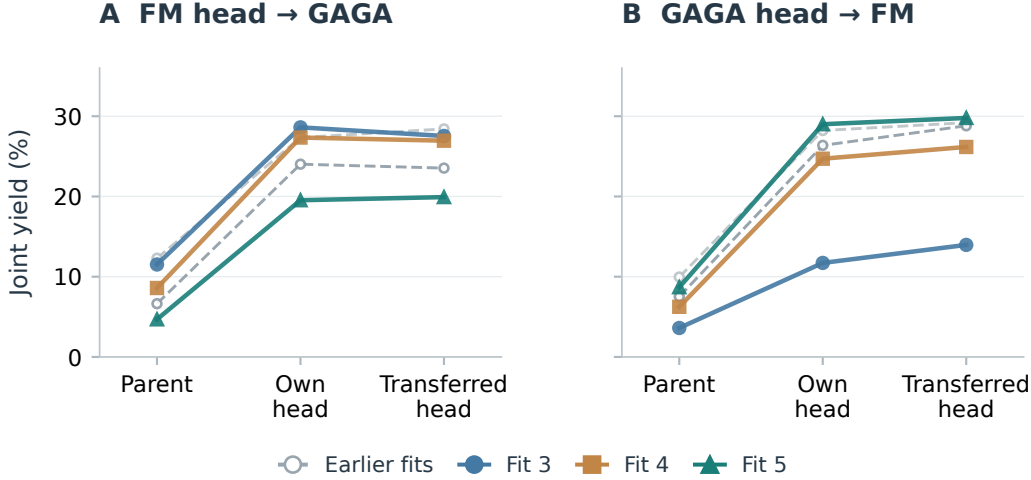


Figure 4: Direct transfer of correction weights. Each line follows one training pair through the uncorrected parent, its own fitted head, and the other model’s head. Dashed open markers show the two earlier pairs; colored filled markers show the three new pairs. Each setting contains 1,024 attempts per pair on the same 64 compositions. The source distribution and target backbone are preserved; the hydrogen readout is disabled.

Direct weight transfer. Across all five pairs, the FM-trained heads raise GAGA joint yield from 8.75 to 25.27%, close to the 25.37% obtained with GAGA-trained heads. In the three new pairs, the gains are 16.02, 18.36 and 15.23 percentage points: a mean of 16.54 points with a crossed fit/composition 95% interval of [12.66, 20.67]. The reverse direction raises FM joint yield from 7.21 to 25.59% across five pairs, compared with 24.00% for its own head. Its three new pairs improve by 17.12 points on average (composition interval [13.61, 20.77]). On the two-fit additional-baseline panel, the same FM heads raise EDM joint yield from 8.15 to 22.22%. These transfers share an EGNN architecture and element vocabulary while changing the generative objective.

Independent training repetitions. For heads fitted to their own parents, the three new FM runs gain 8.11, 18.46 and 20.31 joint-yield points, averaging 15.62 points with crossed interval [8.50, 22.43]. GAGA’s corresponding average is 16.89 points. Across five fits, graph validity changes from 20.12 to 24.86% for FM and from 22.60 to 26.25% for GAGA. Figure 4 retains all five parents, including the weaker FM initialization.

Transferring both designs. A separate experiment retraining EDM and GAGA with the tree-conditioned harmonic noise process and attaches the same frozen FM heads. Joint yield is 0.00% for EDM and 0.05% for GAGA, compared with 8.15% and 9.47% for their two-fit original baselines. Thus this noise adaptation reduces quality in the tested configuration, whereas direct head transfer improves the original generators. Appendix D gives the complete outcomes and training setup for this experiment.

Optional hydrogen readout. Adding the conditional hydrogen flow to the corrected parents gives FM 27.40% graph validity and 26.33% joint yield, and GAGA 26.62% and 25.74%. Their joint-yield

difference is 0.59 points (composition interval $[-1.04, 2.21]$). For the three new FM fits, learned hydrogen placement reduces all-output energy by 19.31 meV/atom relative to the fixed radial rule, with interval $[15.17, 23.67]$ for the reduction. The main comparison, ablation and weight-transfer figure all use outputs before this readout.

6 DISCUSSION

LatentSpring combines a composition-based harmonic source with a correction learned from endpoint forces. The complete flow model improves joint yield over the evaluated baselines, and its correction weights retain their benefit when reused by diffusion generators. The component study locates the source’s additional gain in the corrected flow. Detailed geometry checks show that correction reduces overlaps and force outliers, while heavy-atom connectivity and planarity remain the main structural defects in raw generation.

REPRODUCIBILITY STATEMENT

Code and manifests record seeds, checkpoint hashes, evaluation settings, and all attempted outputs. Training, target preparation, and generation costs are reported separately. Released per-output scores, numerical arrays, and audit summaries reproduce the reported tables and confidence intervals. Tests cover source normalization, symmetry, density gradients, and work identities. Original data and backbone licenses and access requirements apply.

AI USE STATEMENT

OpenAI Codex and Claude Code assisted method and experiment development, code, analysis, figure preparation, and manuscript writing.

REFERENCES

- Christoph Bannwarth, Sebastian Ehlert, and Stefan Grimme. GFN2-xTB — an accurate and broadly parametrized self-consistent tight-binding quantum chemical method with multipole electrostatics and density-dependent dispersion contributions. *Journal of Chemical Theory and Computation*, 15(3):1652–1671, 2019. doi: 10.1021/acs.jctc.8b01176.
- Martin Buttenschoen, Garrett M. Morris, and Charlotte M. Deane. PoseBusters: AI-based docking methods fail to generate physically valid poses or generalise to novel sequences. *Chemical Science*, 15:3130–3139, 2024. doi: 10.1039/D3SC04185A.
- Austin H. Cheng, Chong Sun, and Alán Aspuru-Guzik. Scalable autoregressive 3d molecule generation. *arXiv preprint arXiv:2505.13791*, 2025. doi: 10.48550/arXiv.2505.13791. URL <https://arxiv.org/abs/2505.13791>.
- Gabriele Corso, Vignesh Ram Somnath, Noah Getz, Regina Barzilay, Tommi Jaakkola, and Andreas Krause. Composing unbalanced flows for flexible docking and relaxation. In *International Conference on Learning Representations*, 2025. URL https://proceedings.iclr.cc/paper_files/paper/2025/file/451dbb8f4fca0327ac4e6782786673bf-Paper-Conference.pdf.
- Niclas Dern, Lennart Redl, Sebastian Pfister, Marcel Kollovich, David Lüdke, and Stephan Günnemann. Energy-weighted flow matching: Unlocking continuous normalizing flows for efficient and scalable boltzmann sampling. *arXiv preprint arXiv:2509.03726*, 2025. URL <https://arxiv.org/abs/2509.03726>.
- Leo L. Duan and David B. Dunson. Bayesian spanning tree: Estimating the backbone of the dependence graph. *Journal of Machine Learning Research*, 24(397):1–44, 2023. URL <https://www.jmlr.org/papers/v24/22-0252.html>.
- Ian Dunn and David R. Koes. FlowMol3: Flow matching for 3D de novo small-molecule generation. *Digital Discovery*, 5(5):2052–2066, 2026. doi: 10.1039/D5DD00363F. URL <https://doi.org/10.1039/D5DD00363F>.

- Jasper Marijn Everink. Random spanning tree markov random field priors for bayesian inverse problems in imaging. *arXiv preprint arXiv:2605.18619*, 2026. URL <https://arxiv.org/abs/2605.18619>.
- Majdi Hassan, Nikhil Shenoy, Jungyoon Lee, Hannes Stärk, Stephan Thaler, and Dominique Beaini. ET-Flow: Equivariant flow-matching for molecular conformer generation. In *Advances in Neural Information Processing Systems*, volume 37, 2024. doi: 10.52202/079017-4091. URL https://proceedings.neurips.cc/paper_files/paper/2024/hash/e8bd617e7dd0394ceadf37b4a7773179-Abstract-Conference.html.
- Lars Holdijk, Nithishwer Mouroug Anand, Michael Bronstein, and Max Welling. Learning escorted protocols for multistate free-energy estimation. In *International Conference on Learning Representations*, 2026. URL <https://openreview.net/forum?id=Da8PJXp0js>.
- Emiel Hoogetboom, Víctor Garcia Satorras, Clément Vignac, and Max Welling. Equivariant diffusion for molecule generation in 3D. In *Proceedings of the 39th International Conference on Machine Learning*, volume 162 of *Proceedings of Machine Learning Research*, pp. 8867–8887, 2022. URL <https://proceedings.mlr.press/v162/hoogetboom22a.html>.
- Gerhard Hummer and Attila Szabo. Free energy reconstruction from nonequilibrium single-molecule pulling experiments. *Proceedings of the National Academy of Sciences*, 98(7):3658–3661, 2001. doi: 10.1073/pnas.071034098.
- Gabriel Ilharco, Marco Tulio Ribeiro, Mitchell Wortsman, Suchin Gururangan, Ludwig Schmidt, Hannaneh Hajishirzi, and Ali Farhadi. Editing models with task arithmetic. In *International Conference on Learning Representations*, 2023. URL <https://arxiv.org/abs/2212.04089>.
- Christopher Jarzynski. Equilibrium free energy differences from nonequilibrium measurements: a master equation approach. *Physical Review E*, 56:5018–5035, 1997. doi: 10.1103/PhysRevE.56.5018. URL <https://arxiv.org/abs/cond-mat/9707325>.
- Christopher Jarzynski. Targeted free energy perturbation. *Physical Review E*, 65:046122, 2002. doi: 10.1103/PhysRevE.65.046122. URL <https://arxiv.org/abs/cond-mat/0109461>.
- Junwan Kim, Jiho Park, Seonghu Jeon, and Seungryong Kim. Better source, better flow: Learning condition-dependent source distribution for flow matching. *arXiv preprint arXiv:2602.05951*, 2026. URL <https://arxiv.org/abs/2602.05951>.
- Helen Lai, Tingyu Wang, Hassan Sirelkhatim, Joe Eaton, Howard Huang, Brad Rees, Ola Engkvist, Jon Paul Janet, Xiaoyun Wang, and Alessandro Tibo. Improving protein-ligand complex generation with force field guidance. *Journal of Cheminformatics*, 18:55, 2026. doi: 10.1186/s13321-026-01198-2. URL <https://doi.org/10.1186/s13321-026-01198-2>.
- Daniel S. Levine, Muhammed Shuaibi, Evan Walter Clark Spotte-Smith, Michael G. Taylor, Muhammad R. Hasyim, Kyle Michel, Ilyes Batatia, Gábor Csányi, Misko Dzamba, Peter Eastman, Nathan C. Frey, Xiang Fu, Vahe Gharakhanyan, Aditi S. Krishnapriyan, Joshua A. Rackers, Sanjeev Raja, Ammar Rizvi, Andrew S. Rosen, Zachary Ulissi, Santiago Vargas, C. Lawrence Zitnick, Samuel M. Blau, and Brandon M. Wood. The Open Molecules 2025 (OMol25) dataset, evaluations, and models. *arXiv preprint arXiv:2505.08762*, 2025. URL <https://arxiv.org/abs/2505.08762>.
- Yaron Lipman, Ricky T. Q. Chen, Heli Ben-Hamu, Maximilian Nickel, and Matthew Le. Flow matching for generative modeling. In *International Conference on Learning Representations (ICLR)*, 2023. URL <https://openreview.net/forum?id=PqvMRDCJT9t>.
- Jiarui Lu, Zuobai Zhang, Bozitao Zhong, Chence Shi, and Jian Tang. Fusing neural and physical: Augment protein conformation sampling with tractable simulations. In *GEM Workshop at ICLR*, 2024. URL <https://arxiv.org/abs/2402.10433>.
- Marina Meila and Tommi Jaakkola. Tractable bayesian learning of tree belief networks. Technical Report CMU-RI-TR-00-15, Carnegie Mellon University, 2000. URL <https://publications.ri.cmu.edu/tractable-bayesian-learning-of-tree-belief-networks>.

- Laurence Illing Midgley, Vincent Stimper, Gregor N. C. Simm, Bernhard Schölkopf, and José Miguel Hernández-Lobato. Flow annealed importance sampling bootstrap. In *International Conference on Learning Representations (ICLR)*, 2023. URL <https://arxiv.org/abs/2208.01893>. arXiv:2208.01893.
- Jingxiang Qu, Wenhan Gao, Ruichen Xu, and Yi Liu. GAGA: Gaussianity-aware gaussian approximation for efficient 3d molecular generation. In *International Conference on Learning Representations (ICLR)*, 2026. URL https://proceedings.iclr.cc/paper_files/paper/2026/hash/08de3c1eb4adc7ac3c1949048422d895-Abstract-Conference.html.
- Víctor García Satorras, Emiel Hoogetboom, and Max Welling. E(n) equivariant graph neural networks. In *Proceedings of the 38th International Conference on Machine Learning*, volume 139, pp. 9323–9332. PMLR, 2021. URL <https://proceedings.mlr.press/v139/satorras21a.html>.
- Chence Shi, Shitong Luo, Minkai Xu, and Jian Tang. Learning gradient fields for molecular conformation generation. In *Proceedings of the 38th International Conference on Machine Learning*, volume 139, pp. 9558–9568, 2021. URL <https://proceedings.mlr.press/v139/shi21b.html>.
- Hannes Stark, Bowen Jing, Regina Barzilay, and Tommi Jaakkola. Harmonic self-conditioned flow matching for joint multi-ligand docking and binding site design. In *Proceedings of the 41st International Conference on Machine Learning*, volume 235, pp. 46468–46494. PMLR, 2024. URL <https://proceedings.mlr.press/v235/stark24a.html>.
- Suriyanarayanan Vaikuntanathan and Christopher Jarzynski. Escorted free energy simulations: Improving convergence by reducing dissipation. *Physical Review Letters*, 100:190601, 2008. doi: 10.1103/PhysRevLett.100.190601. URL <https://arxiv.org/abs/0804.3055>.
- Vikram Voleti, Adam Oberman, and Christopher Pal. Score-based denoising diffusion with non-isotropic gaussian noise models. *arXiv preprint arXiv:2210.12254*, 2022. URL <https://arxiv.org/abs/2210.12254>.
- Cong Wang, Hanxin Zhu, Jiayi Luo, Yonglin Tian, Xiaoqian Cheng, Peiyan Tu, Xin Jin, Long Chen, and Zhibo Chen. Physics-informed video generation via mixture-of-experts latent alignment. *arXiv preprint arXiv:2606.04737*, 2026a. URL <https://arxiv.org/abs/2606.04737>.
- Yan Wang, Lihao Wang, Yuning Shen, Yiqun Wang, Huizhuo Yuan, Yue Wu, and Quanquan Gu. Protein conformation generation via force-guided SE(3) diffusion models. In *Proceedings of the 41st International Conference on Machine Learning*, volume 235, pp. 56835–56859. PMLR, 2024. URL <https://proceedings.mlr.press/v235/wang24cv.html>.
- Zifan Wang, Alice Harting, Matthieu Barreau, Michael M. Zavlanos, and Karl H. Johansson. Source-guided flow matching. In *International Conference on Learning Representations*, 2026b. URL https://proceedings.iclr.cc/paper_files/paper/2026/hash/adedc33539eafdf3d2c14a5e0032db6e-Abstract-Conference.html.
- Minkai Xu, Lantao Yu, Yang Song, Chence Shi, Stefano Ermon, and Jian Tang. GeoDiff: A geometric diffusion model for molecular conformation generation. In *International Conference on Learning Representations*, 2022. URL <https://arxiv.org/abs/2203.02923>.
- Zhijian Zhou, Junyi An, Zongkai Liu, Yunfei Shi, Xuan Zhang, Fenglei Cao, Chao Qu, and Yuan Qi. Guiding diffusion models with reinforcement learning for stable molecule generation. *arXiv preprint arXiv:2508.16521*, 2025. URL <https://arxiv.org/abs/2508.16521>.

A READER’S GUIDE TO THE NOTATION

LatentSpring first samples starting coordinates from atomic composition, then learns a flow to reference structures, and incorporates physical feedback through paired fine-tuning or an explicit velocity correction. This appendix collects the notation used in those stages.

Inputs and outputs. The condition c fixes the atom identities, charge, and spin. Random starting coordinates X_0 provide variation between generated structures. Each network evaluation takes current coordinates X_t , flow time t , and c and predicts one velocity vector per atom. Self-conditioning uses a first prediction to form a provisional geometry for a second pass. Integrating the velocities and adding the specified terminal noise gives the final coordinate matrix X_{gen} . The learned-correction variant sets this terminal noise to zero. The input atom identities and electronic conditions do not change. Bonds are inferred separately for evaluation.

Symbol	Meaning
N, z_i, Q, S, c	Atom count, atomic numbers, total charge, and spin multiplicity. The condition c collects these inputs.
$X, x_i, \mathcal{H}_N$	Atomic coordinate matrix, the position of atom i , and the space where the mean atomic position is zero. Coordinates are in angstroms.
T, a_{ij}, u_i	Auxiliary spatial tree, its edge affinities, and the atom-dependent factors defining those affinities. These are source variables.
$\ell_{ij}, \sigma_{ij}, h_{ij}$	Characteristic separation from covalent radii, Gaussian displacement scale, and the probability density of one edge vector.
$\tau(a), q_0(X c)$	Sum of tree weights and the normalized distribution of starting coordinates after averaging over trees.
$X_0, X_1, X_t, X_{\text{gen}}, t$	Starting, reference, intermediate and returned coordinates; t labels generation progress, not physical time.
V^*, v_θ, D_θ	Target velocity, predicted velocity, and the coordinate-shaped network head whose centered increment $D_\theta - X_t$ is trained as the velocity.
$v^{(0)}, v^{(1)}, \hat{X}_1$	First and second velocity predictions, and the provisional endpoint passed between them during self-conditioning.
v_{base}, H_t	Frozen parent velocity after both self-conditioning passes, and its extrapolated endpoint used to construct physical training targets.
$A_\phi, \phi, v_{\text{cap}}$	Added velocity, correction-network parameters, and its instantaneous velocity-bound scale.
$\mathbb{E}, \ \cdot\ _F^2$	Average over the specified random draws, and the sum of squared entries of a coordinate or force matrix.
$\theta_{\text{base}}, \theta_{\text{physical}}, \theta_{\text{replay}}, \theta_{\text{final}}$	Base weights, weights after physical or replay fine-tuning, and the final generation weights.
$A, q_A, \sigma_A, \delta_A$	Generated anchor, local Gaussian reference, perturbation width, and force-guided shift.
E_+, F_+	Mirror-symmetrized teacher energy and force, in eV and eV/Å. The force is the negative coordinate gradient of this energy.
$k_B T_{\text{phys}}, W_A$	Thermal energy at the teacher temperature and generalized work used to weight local candidate structures.
Φ, J, Z	Coordinate map, its local volume-change factor, and a target-density normalizing integral.

Table 1: Notation for the source, flow model, and physical update. The tree T and teacher temperature T_{phys} are different quantities.

Structural and physical metrics. Geometric validity requires one connected contact structure with no excessive atom overlap. Chemical-graph validity additionally requires a chemically consistent bond graph inferred from the coordinates. Two paired outputs are *jointly valid* when both pass these checks; they need not have the same inferred graph. This differs from the physical teacher’s *same-graph* restriction, which preserves the anchor’s labeled bond orders, formal charges, and radical counts.

We compare energy within the same atomic composition and report changes per atom. Force RMS is $[N^{-1} \sum_{i=1}^N \|F_i\|_2^2]^{1/2}$, the root mean squared magnitude of the atomic force vectors. Lower force RMS means smaller residual forces. Lower energy means greater energetic preference under the chosen potential. These are different measurements from graph validity.

B IMPLEMENTATION AND ANALYSIS OF THE MAIN EXPERIMENTS

LatentSpring configuration. The main model uses the shared EGNN with four message-passing layers, 256 hidden features, and 2,381,566 backbone parameters. It conditions on fixed atom identities and uses the harmonic-mixture source and distance self-conditioning. Training uses AdamW, a learning rate of 10^{-4} , batch size 32, and 15,000 updates with two supervised forward passes per example. The final exponential moving average is used, with maximum decay 0.999. The source edge log-width is 0.2. The implementation supports up to 200 atoms; the main evaluation contains 17–40 atoms. The shared training set contains 20,000 OMol25 structures with 8–64 atoms, as specified in Appendix K.

The physical head contains 7,106 parameters, a 16-dimensional atom embedding, and a pair network with hidden width 64. Its velocity bound is $2p^2 \text{ \AA}$ per unit progress. For each fit, 256 native trajectories on 128 training compositions provide 512 late provisional endpoints. Mirror-paired eSEN evaluations use 1,024 force queries. Of the 128 compositions, 96 fit the head and 32 measure target-prediction error. Head training uses AdamW at learning rate $3 \cdot 10^{-4}$ for 20,000 updates. Force targets use $T_{\text{phys}} = 300 \text{ K}$, a maximum local width of $0.03 \text{ \AA}$, and a maximum molecular displacement norm of $0.1 \text{ \AA}$. The correction strength is fixed at four before the five-fit confirmation and the additional-baseline experiment.

Generation uses 32 midpoint steps with two self-conditioning passes per field evaluation: 128 backbone calls and 64 small-head calls per output. No terminal coordinate perturbation, hydrogen readout, physical query, or geometry optimization is used in Figure 2. Head training and teacher queries are additional to backbone training. The main comparison matches backbone training-example forward counts and backbone sampling calls; it does not equate total physical supervision, FLOPs, or wall time. Archived zero-strength controls execute unused head calls to verify sampler behavior; those diagnostic calls are not part of an unaugmented baseline’s required inference procedure.

Evaluation and repetition counts. The 64-composition panel contains 20 compositions with 17–28 atoms and 44 with 29–40 atoms, all absent from the processed training and validation corpora. Figure 2 uses five fits for LatentSpring and GAGA, and two archived fits for Gaussian FM and EDM. Each fit produces 1,024 outputs. The extra three FM/GAGA fits were trained after freezing the method; Appendix N.2 separates these from the earlier two fits that informed development. The Gaussian FM and EDM evaluations reuse the same panel in a later experiment with settings fixed before generating their outputs. No checkpoint or output is selected by its evaluation energy.

Four-setting component analysis. Figure 3 uses the two archived, one-pass Gaussian and harmonic FM parents from Appendix K. Both omit self-conditioning, unlike the main LatentSpring row. Within each fit, we apply the identical frozen head previously learned on the distance-self-conditioned FM, with strength four and no additional target-specific training. Each output uses 128 backbone calls and, when active, 128 head calls. The parent and corrected trajectory share source draws and sampler randomness. This compares the two sources under one fixed learned correction. It is not a four-cell retraining of the final self-conditioned model. The four-cell contrasts were specified after the frozen transfer experiment and are reported as a follow-up analysis.

Table 2: Joint yield in the four-setting analysis, by training fit (percent).

Source	Physical correction	Fit 1	Fit 2
Gaussian	No	5.57	6.93
Harmonic	No	6.93	7.52
Gaussian	Yes	19.73	20.12
Harmonic	Yes	23.24	25.29

Paired bootstrap intervals use 20,000 resamples. Resampling both fit indices and compositions gives a nominal 95% interval of $[1.56, 7.13]$ percentage points for the combination relative to Gaussian FM with correction, and $[13.09, 21.19]$ relative to harmonic FM without correction. The source-only joint-yield change is $+0.98$ points, with interval $[-0.68, 2.54]$. The four-cell table therefore supports the combined method while the smaller source-only physical-quality difference remains uncertain.

Three initially unsuccessful GFN2 calculations in the Gaussian arms remain in the denominator; they are not replaced or counted as joint successes.

Structured diffusion in the combined-design transfer. In Equation 7, B_T is the centered tree embedding multiplied by the edge standard deviations, so C_T has rank $N - 1$. If D_T takes oriented edge differences divided by their standard deviations, the training loss is $\|D_T(\hat{n} - B_T\Xi)\|_F^2/[3(N - 1)]$, where $\hat{n}$ is the denoiser’s predicted coordinate noise. This is the inverse-covariance weighted reconstruction error on the centered subspace. The existing second invariant edge channel receives $\log[1 + 3(C_{T,ii} + C_{T,jj} - 2C_{T,ij})]$. The covariance adds no learned parameters and contains no observed chemical bonds.

EDM starts from $B_T\Xi$. At GAGA’s selected truncation, the starting Gaussian conditional on T has covariance $\alpha_t^2 v P + \sigma_t^2 C_T$. Here v is the original training-only coordinate variance and $P = I - \mathbf{1}\mathbf{1}^\top/N$ removes translation. Thus GAGA retains its data-variance approximation while using harmonic corruption noise. Both structured backbones use the paired original random initializations, the original 20,000 references and batch schedules, and 30,000 one-pass updates. The two FM physical heads remain frozen. The target models are retrained, so this experiment transfers the two designs; only the physical head is reused without retraining. Appendix D reports the completed generation and physical-score results.

Released evidence. The additional experiment evaluates all three target baselines (Gaussian FM, harmonic FM, and EDM), with and without the same transferred FM head, in both archived fits. It adds 12,288 generated outputs and fixed-coordinate GFN2 attempts, with no new training or eSEN queries. Raw-coordinate hashes, independently parsed physical logs, all-output score arrays, and numerical summaries are archived in the released evidence. A separate 48-output replay check verifies that zero-strength correction reproduces native sampling. The main-table renderer reads the audited summaries directly; the four-setting analysis script reconstructs its contrasts from the released score arrays.

C CHEMICAL GEOMETRY AND ILLUSTRATIVE STRUCTURES

We supplement graph inference and the GFN2 force criterion with geometric checks from PoseBusters 0.4.4 (Buttenschoen et al., 2024). The checks use distance bounds for heavy-atom bond lengths and angles, internal steric distances, planar aromatic rings and double bonds, and nonplanarity of saturated six-membered rings. We use the package’s molecule-configuration tolerances: 25% for bond and angle bounds, 30% for internal distances, 0.25 Å for planar groups, and 0.05 Å for nonplanar rings. A structure must also have zero assigned radical electrons. The check is a geometric subset of PoseBusters; it omits conformer ensemble energy ratios and reference-identity comparisons.

Every generated coordinate remains fixed. We evaluate all 5,120 attempts per setting, using the same inferred graph as the original benchmark. This includes outputs with failed graph inference in the denominator. Table 3 reports the geometric subset alone and its conjunction with the original GFN2 force threshold of 5 eV/Å.

Table 3: Additional geometric checks on the five-fit panel. Percentages use all 5,120 attempts per row. “Geometry” includes graph validity, zero assigned radicals, and the geometric tests described above.

Method	Graph validity	Geometry	Geometry + force
FM parent	20.12	3.48	1.64
FM + correction	24.86	4.61	4.43
GAGA parent	22.60	4.71	2.40
GAGA + correction	26.25	6.64	6.45

For FM, correction reduces contact-disconnected outputs from 3,092 to 2,702, heavy-atom-disconnected outputs from 2,527 to 2,325, and severe overlaps from 126 to nine. Among the 1,273 graph-valid corrected outputs, 674 fail a planarity check, 340 fail a bond-angle check, 237 fail a bond-length check, and 296 fail the internal-distance check. These categories overlap. On all 512 prepared validation references, 470 pass the same closed-shell geometric screen. The comparison

identifies connectivity and detailed geometry as remaining generation errors even after the force criterion improves.

The molecular illustration. Figure 1 uses an additional validation-composition panel chosen from reference structures before generation. The selection takes the first four distinct compositions, in original validation-row order, whose references have 17–28 atoms, CHONF elements, uncharged atoms, five- or six-membered rings, and pass the geometric checks. Each composition receives 32 raw outputs from the unchanged main-model fit 1, with correction strength four and the hydrogen readout disabled. All 128 outputs are preserved and evaluated: 79 pass graph inference and 15 pass the closed-shell geometric screen.

The displayed $\text{C}_9\text{H}_9\text{NO}_2$ structure is condition index 2, draw 27 (zero-based), with inferred connectivity $\text{N}=\text{Cc1ccccclC1OC1O}$. It is the first output in condition/draw order with an aromatic ring, uncharged atoms, zero assigned radicals, passing geometry checks, and force RMS at most 5 eV/Å. Its force RMS is 2.886 eV/Å. An earlier illustration screen that excluded every three- and four-membered ring had no eligible output; the displayed example contains a saturated epoxide. This panel supplies an illustration and is separate from the 64-composition benchmark.

We replay the source random numbers exactly and use the stored output coordinates. The 3D source and output share camera and scale. The 2D inset depicts the inferred connectivity, with hydrogens implicit in the usual chemical notation. Bond orders are rendered from an explicit Kekulé representation, and the PyMOL round-trip check verifies atom identities, bond orders and coordinate precision. No molecular geometry optimization is used in preparing the illustration.

D COMBINED SOURCE AND CORRECTION TRANSFER TO DIFFUSION

Extending the source to diffusion. To transfer both designs, we use the harmonic mixture as structured diffusion noise. For a sampled tree T , let B_T map independent standard Gaussian edge vectors to centered coordinates with the scales in Equation 2. Its coordinate covariance is $C_T = B_T B_T^\top$. A forward diffusion state is

$$X_t = \alpha_t X_{\text{clean}} + \sigma_t B_T \Xi, \quad \Xi \in \mathbb{R}^{(N-1) \times 3}, \quad \Xi_{ij} \sim \mathcal{N}(0, 1). \quad (7)$$

The tree stays fixed along a trajectory, and reverse-sampling noise uses the same covariance. The denoiser receives invariant pair variances from C_T and predicts the noise $B_T \Xi$ in coordinate units. This follows the established non-isotropic Gaussian diffusion construction (Voleti et al., 2022). The target backbone is trained for this noise process; the FM physical head is then reused through the same endpoint correction. Appendix B gives the covariance conditioning and GAGA initialization.

The combined-design experiment trains tree-conditioned diffusion models using Equation 7, then attaches the frozen FM correction head. EDM and GAGA each use two original initialization seeds, the same 20,000 OMol25 rows and batch schedules as their baselines, and 30,000 updates per fit. The source covariance is held fixed along each trajectory and supplied through an invariant edge channel. Every model uses its final EMA checkpoint, 128 backbone evaluations per sample, correction strength four, and no hydrogen readout. The baselines reuse their audited original outputs on the same 64 compositions. Each setting contains 2,048 attempts.

Table 4: Original diffusion baselines and combined-design adaptations (percent).

Target	Setting	Graph validity	Joint yield
EDM	Original	22.27	8.15
EDM	Harmonic noise + FM head	0.00	0.00
GAGA	Original	24.22	9.47
GAGA	Harmonic noise + FM head	0.24	0.05

The adaptation reduces generation quality in both target families. Its two GAGA fits produce one and four graph-valid outputs, respectively; only one also passes the force threshold. The EDM fits produce no graph-valid outputs. There are 36 unsuccessful GFN2 calculations for EDM and 21 for GAGA; all remain in the denominators. The experiment contributes 120,000 backbone updates, 3.84 million training-example forwards, 4,096 generated trajectories and GFN2 attempts, with no new

head training or force-teacher queries. The released protocol, training logs, arrays and audit retain every fit.

E EARLIER SOURCE-STUDY SETUP

Task and training. The earlier source experiments adapt a pretrained molecular flow to neutral singlet organic compositions. Every composition contains C and H, with elements drawn from H/B/C/N/O/F/Si/P/S/Cl/Br/I. Training covers 8–40 atoms and evaluation extends to 64; the implementation supports up to 200 atoms. The initial two fine-tuning runs start from the same pretrained checkpoint. Each uses 3,000 distinct OMol25 structures and 3,000 optimizer updates. Within a run, source variants share the training data and initial weights. The network has 5,904,369 parameters. Learning rates are $2 \cdot 10^{-5}$ for the backbone and $3 \cdot 10^{-4}$ for self-conditioning modules. Three additional source runs repeat these two data blocks with new random seeds and report all 1,000-, 3,000-, and 6,000-update checkpoints.

All source variants use the same filtered training subset. In addition to the chemical-graph checks, selection requires a spanning scaffold with bounded edge lengths and atom degrees. This scaffold is a data-selection filter, not an input to the model. Appendix F specifies it.

Earlier FlowMol output and sampler. The FlowMol implementation uses a coordinate-shaped head D_θ and interprets the centered increment $D_\theta - X_t$ as velocity. In its original midpoint scheme, the first evaluation uses the current coordinates and the interval’s midpoint time; the second uses the half-step coordinates at that same time. The source-only comparisons add centered Gaussian noise of standard deviation 0.025 Å after integration. The learned-correction model omits this terminal noise. These implementation details differ from the shared-EGNN main comparison.

Evaluation sets. The source study uses four composition sets: 12 for development, 10 for an additional test, 24 for a further test, and 64 for a broader size comparison. Exact atom-count checks confirm that the latter three sets are absent from both processed OMol25 corpora. Their reference coordinates are used only to determine eligibility, not for training. Initial comparisons reuse two fine-tuned models, and the wider source study adds three training repetitions. Shared-EGNN experiments train from scratch; the final correction study includes five paired initializations. Physical studies compare models on common compositions and identify the panel alongside each result. Selection and source-control details are in Appendix G.

Metrics and uncertainty. The 12- and 10-composition sets use 64 samples per composition, and the 24-composition set uses 32. Geometric validity requires a connected contact graph and no severe atom overlap, with distance thresholds of 1.25 and 0.6 times the summed covalent radii, respectively. Chemical-graph validity additionally requires RDKit to infer a connected, sanitized bond graph with the requested total charge. We divide valid outputs by *all attempts*, including failures and validator exceptions. Graph diversity counts distinct inferred connectivities within each composition. Timings include the full generation procedure. Appendix F gives the detailed checks.

Unless stated otherwise, 95% confidence intervals come from paired bootstrap resampling of generation draws, with the trained models and compositions held fixed. We also resample compositions to describe variation across the evaluation set. Physical scores are computed on saved raw coordinates, using the stated charge and spin and approximate energy models.

F DATA SELECTION AND STRUCTURAL EVALUATION

We traverse the processed OMol25 training corpus in a fixed shuffled order and exclude evaluation compositions. The first 3,000 eligible structures form the training set for each fine-tuning run. Eligibility requires the atom-count, element, and electronic-state conditions of the source study in Appendix E, the structural checks below, and a geometric scaffold filter.

Training-only scaffold filter. For each atom pair, we divide its distance by the sum of the atoms’ covalent radii. Only pairs with normalized distance strictly between 0.6501 and 1.1999 can be scaffold edges. Kruskal’s algorithm builds a minimum normalized-distance spanning tree by adding

short edges without creating cycles. We accept a structure only if the tree is connected and its atom degrees are at most 1 for H/F/Cl/Br/I, 4 for B/C/N/Si, 3 for O, and 6 for P/S. This tree filters training coordinates; it is not the random tree used to sample the model’s source.

Inferring and checking bonds. We use RDKit 2025.03.2 for bond inference and chemical-consistency checks. After testing contact connectivity and atom overlap, `DetermineBonds` assigns bonds at the requested total charge with `covFactor=1.25`, `useVdw=True`, `allowChargedFragments=True`, and `embedChiral=False`. RDKit sanitization checks valence and related graph rules. An output passes only if sanitization succeeds, formal charges sum to the requested charge, and the inferred graph has one connected component. To measure graph diversity, we remove stereochemical tags and explicit hydrogens and count distinct canonical SMILES, which encode molecular connectivity. Validator errors and rejected structures remain in the generation-attempt denominator.

The 24-composition confirmation set. Before this evaluation, we freeze the Gaussian, harmonic, and two physically updated harmonic models and designate the force-only update as the primary physical model. We start from 256 distinct compositions outside the earlier candidate sets; checking their reference structures leaves 121 eligible compositions. Fixed metadata ranks select eight from each of the 17–20, 21–24, and 25–28 atom ranges. None of the 24 appears in either processed OMol25 training or validation corpus. Reference coordinates are used only to determine eligibility. With 32 samples per composition in each run, the four models produce 6,144 raw outputs.

G SOURCE-DEVELOPMENT AND COVARIANCE COMPARISONS

Composition sets. Checking reference structures leaves 31 eligible compositions out of 78 that pass the metadata conditions. Of these, 21 had appeared in earlier development experiments. We use a fixed subset of 12 for source development and all 10 remaining compositions for a subsequent test. These 10 have 17–28 atoms and are absent from 3,902,107 processed training rows and 39,415 validation rows, verified by exact atomic-count vectors. Reference geometries only determine eligibility. Both sets use the same two fine-tuning runs; the further 24-composition test is described in Section L.2.

On the development set, the harmonic source improves graph validity over the isotropic Gaussian by 5.86 percentage points (paired 95% CI [2.67,8.98]; composition-bootstrap interval [0.52,12.17]). Both runs improve, with a larger gain in run 1. Harmonic also exceeds the narrow shell-tree source by 4.23 points; the shell-minus-harmonic interval is [-7.29,-1.11].

On the 10-composition test, harmonic improves graph validity from 52.73% to 61.56%, a gain of 8.83 points (paired 95% CI [5.08,12.50]; composition-bootstrap interval [4.84,12.66]). Geometric validity improves by 9.69 points [6.33,12.89]. Every graph-valid output has distinct inferred connectivity within its composition, and no validator exceptions occur. Generation time is nearly unchanged: harmonic/Gaussian take 175.73/175.43 seconds for 640 samples in run 1 and 178.20/177.74 seconds in run 2.

Control for source covariance. To test whether the source benefit is explained by its covariance, we compare against a single Gaussian with approximately the same covariance as the harmonic mixture. We estimate that covariance from 128 trees and average over permutations of identical atom species. An independent 8,192-tree estimate gives a relative Frobenius error of 1.85–6.25% across the 22 compositions. We match the training data, optimizer, network, and sampling budgets. This follow-up compares the new Gaussian models with the already frozen harmonic checkpoints.

Harmonic improves graph validity over this Gaussian by 11.65 percentage points on development (paired 95% interval [8.59,14.78]; descriptive composition interval [8.07,15.43]) and 15.39 points on the additional set (paired [11.64,19.06]; composition [11.95,18.98]). The difference is positive in both runs on both sets. The covariance-matched Gaussian also performs worse than the isotropic Gaussian. These results support a benefit beyond this calibrated second-moment approximation, consistent with a contribution from the mixture structure. The comparison adds 2,816 outputs while holding parameter count, initial self-conditioning weights, training structures, and optimization fixed.

Table 5: Valid outputs divided by all generation attempts. All variants use the same geometry self-conditioning backbone and 128 network forward passes per output. The additional set evaluates models frozen after the development comparison.

Set	Run	Gaussian	Shell tree	Harmonic tree
Development (12)	1	357/768	417/768	436/768
	2	377/768	342/768	388/768
Additional (10)	1	326/640	—	413/640
	2	349/640	—	375/640

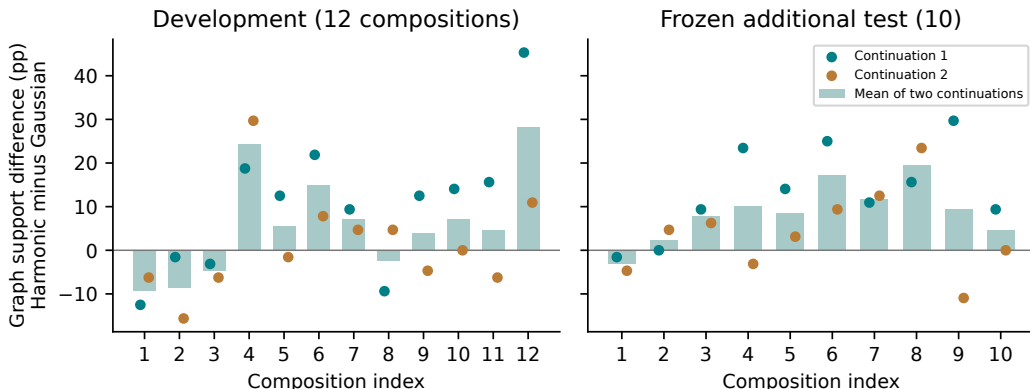


Figure 5: Harmonic-minus-Gaussian graph-validity differences for each composition. Points show the two individual fine-tuning runs, and bars show their mean. Gains vary across compositions and runs. Both evaluation sets use the same trained models.

H VALIDITY BY MOLECULAR SIZE

The 64-composition evaluation set contains 16 compositions in each size range. The table separates the pooled source comparison by atom count.

H.1 REPEATED SOURCE TRAINING AND BUDGET DEPENDENCE

We add three paired fine-tuning runs from the same pretrained FlowMol checkpoint. Runs 3 and 5 use the first 3,000-row training block; run 4 uses the second. Each pair shares initial weights, reference order, and optimizer settings, and changes only the source. At 6,000 updates, the same 3,000 rows have been visited twice. These are five stochastic fine-tuning runs on two fixed data blocks, not five independent pretraining runs or an isolated dataset-size experiment.

Every new run is evaluated at 1,000, 3,000, and 6,000 updates on the existing 64-composition panel. The first and last checkpoints use 16 draws per composition; the middle checkpoint uses 32, matching the original runs. All checkpoints and all runs are reported. No checkpoint was selected using these outcomes.

At 3,000 updates, the five-run mean is +3.89 points and the sample standard deviation across runs is 5.27 points. The composition-bootstrap interval is [2.52, 5.33], conditional on these fitted models; it does not describe uncertainty over future training runs. For the three new runs, the corresponding intervals are [-4.04, 0.68], [0.36, 3.52], and [1.73, 6.25] at the three budgets. The source advantage depends on training duration and varies across runs. This study adds 36,000 optimizer updates and 24,576 raw outputs, with no physical-oracle queries.

Table 6: Covariance-matched Gaussian versus harmonic source. Entries count valid outputs over all attempts. Training and inference budgets are matched; the Gaussian uses the calibrated covariance estimate from 128 trees.

Set	Run	Covariance Gaussian	Harmonic tree
Development	1	332/768	436/768
	2	313/768	388/768
Additional	1	286/640	413/640
	2	305/640	375/640

Table 7: Graph validity on the 64-composition set, measured before geometry optimization. Each size range contains 1,024 attempts per method across the two fine-tuning runs. Rates are percentages; differences are percentage points.

Atoms	Gaussian	Harmonic	Difference
17–28	49.12	55.08	+5.96
29–40	24.02	38.28	+14.26
41–52	16.50	21.29	+4.79
53–64	9.77	12.40	+2.64
All 64 compositions	24.85	31.76	+6.91

I ADDITIONAL SOURCE RESULTS

I.1 EFFECT OF THE LATENT SPRING SOURCE

Replacing the isotropic Gaussian source with the harmonic source improves chemical-graph validity by 5.86 percentage points on the 12 development compositions, 8.83 points on the 10 additional compositions, and 8.20 points on the 24-composition set. These comparisons hold the backbone, initialization, and training data fixed. A Gaussian with approximately matched covariance also performs worse. This supports a benefit beyond the tested covariance approximation, without isolating every property of the mixture. Appendix G reports the source, shell-tree, and covariance controls.

I.2 GENERALIZATION ACROSS MOLECULAR SIZES

We test whether the source advantage persists for larger molecules without further training. The evaluation set contains 64 new compositions, with 16 in each of four size ranges spanning 17–64 atoms. None appears in either processed OMol25 corpus or in the earlier evaluation sets. We select them by fixed metadata ranks after the same reference-validity checks, then generate 32 samples per composition in each fine-tuning run.

Across all 64 compositions, graph validity is 24.85% with the Gaussian source and 31.76% with the harmonic source. The pooled harmonic result is higher in every size range (Appendix H). The overall gain is 6.91 percentage points, with a 95% interval of [4.74, 9.03] from resampling compositions within size ranges. The two fine-tuning runs improve by 11.18 and 2.64 points, respectively. Validity decreases as molecules become larger, but the aggregate source advantage remains positive beyond the original 17–28-atom test range.

Three further fine-tuning runs bring the 3,000-update comparison to five runs. The average improvement is 3.89 points, positive in four runs, with a between-run standard deviation of 5.27 points. For the three new runs, the average advantage changes from −1.66 points at 1,000 updates to +1.87 at 3,000 and +3.94 at 6,000. All three favor the harmonic source at 6,000 updates. Appendix H.1 reports every run and training budget.

Table 8: Harmonic-minus-Gaussian graph validity in percentage points. Dashes mark training budgets not run for the original two models.

Fine-tuning run	1,000 updates	3,000 updates	6,000 updates
1	—	+11.18	—
2	—	+2.64	—
3	−1.37	+5.66	+5.96
4	−0.10	+3.32	+3.91
5	−3.52	−3.37	+1.95
Mean, new runs 3–5	−1.66	+1.87	+3.94
Mean, all available runs	−1.66	+3.89	+3.94

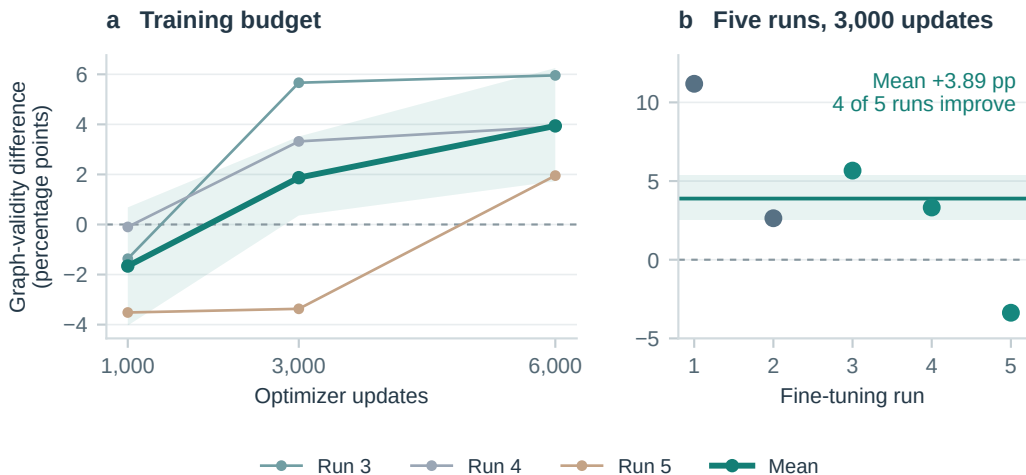


Figure 6: Source effects across training budgets and fine-tuning runs. Thin lines show all three new runs; the thick line is their mean. The right panel includes all five runs at 3,000 updates. Shading and the horizontal bar show 95% composition-bootstrap intervals conditional on the fitted models.

I.3 INDEPENDENT BACKBONE AND GENERATOR OBJECTIVES

We train Gaussian FM, harmonic FM, position-conditional EDM, and GAGA (Hooeboom et al., 2022; Qu et al., 2026) from scratch on an independent EGNN. All four use the same 20,000 OMol25 structures, architecture, and initial weights within each of two repetitions, with 30,000 updates at batch size 32. This comparison omits self-conditioning and physical adaptation. The two FM models differ only in source distribution; their probability paths and atom pairing are identical. Appendix K gives the protocol and GAGA configuration.

The harmonic source improves FM graph validity by 1.34 percentage points, with a composition-bootstrap 95% interval of [0.12, 2.52]. This supports a source benefit without shared pretraining history. EDM and GAGA achieve higher validity in this setting. Unlike the Gaussian-harmonic FM comparison, those comparisons also change the probability path, training objective, and sampler.

Distance self-conditioning provides the FM parent for the physical-head comparison, with matched training forward-pass budgets. The original feedback and two-fit correction experiments are reported in Appendices K.1 and N.

A further 16-composition comparison evaluates complete systems with their own architectures and training histories. The 5.90-million-parameter FlowMol backbone reaches 38.87% GFN2 joint yield, increasing to 46.88% with learned force correction. GAGA reaches 9.77% at 128 calls and 12.11% at 651 calls, using a 2.38-million-parameter backbone trained from scratch. Their inherited training and physical supervision differ. Appendix O reports all variants, preparation costs, and the matched-size

and training-duration controls. These complete-system results complement the architecture-controlled experiments above.

Independent physical evaluation of weight updates. On 24 new compositions, paired physical training lowers energy and force RMS under both the eSEN teacher and the independent GFN2 evaluator. Graph validity remains similar. Appendix L.2 gives all-attempt counts, paired valid-output comparisons, and the complete-work control.

Learning physical feedback in weights. The paired update also lowers energy and force relative to a direct update of the same parameter norm. Both fitted models favor the paired direction. Appendix L.1 reports this control, and Appendix L.3 compares physical adaptation under both source distributions.

Component controls. At matched training-forward budgets, self-conditioning improves validity by 12.43 points ([9.70, 15.23]). Removing the force displacement raises energy by 0.01939 eV/atom on paired graph-valid outputs. Appendix W reports both runs, validity, force scores, and the update-matched comparison.

J ADDITIONAL GENERATOR EVALUATIONS

J.1 AN INDEPENDENTLY ADAPTED DIFFUSION BASELINE

We also adapt the official EGNN/EDM backbone (Hooeboom et al., 2022) to composition-conditioned coordinate generation. Atom categories are fixed conditions during training and sampling, while positions follow EDM’s variance-preserving diffusion schedule. Starting from its GEOM-pretrained checkpoint, we train for 30,000 updates on the same 3,000 OMol25 structures used in each source-model run, retaining the original learning rate of 10^{-4} . We evaluate all 24 new compositions with both a 128-call sampler and the full 1,001-call sampler, including the final observation step.

This baseline uses the same task and adaptation data but a different pretrained architecture and objective. EDM has 2.38 million parameters and one denoiser pass per update; LatentSpring has 5.90 million parameters and two passes. The longer EDM fine-tuning runs take 481.51 and 463.61 seconds, compared with about 570–590 seconds for the source-model runs. For 768 outputs, EDM’s 128-call sampler takes 48.54 and 47.94 seconds; its 1,001-call sampler takes 377.65 and 374.12 seconds. Matching network-call counts is therefore not the same as matching computational cost. Pretraining histories also differ, and the composition-overlap checks cover only the processed OMol25 corpora. We use the matched Gaussian–harmonic experiments, rather than this baseline, to isolate the effect of source choice.

K INDEPENDENT-BACKBONE COMPARISON

This experiment tests the source without pretraining on a second architecture. All methods use the official EDM equivariant graph neural network (EGNN) (Hooeboom et al., 2022), with four message-passing layers, 256 hidden features, and 2,381,566 parameters. We repeat training from two independent initializations. Within each repetition, all methods start from identical weights and receive the same minibatches. Atom types are fixed inputs throughout training and generation.

Data. The shared data contain 20,000 OMol25 training structures from 8,323 compositions and 512 internal validation structures from 299 other compositions. Both sets contain neutral organic singlets with 8–64 atoms. A hash of composition assigns structures to training or internal validation. Selection requires the geometric and chemical-graph checks, but not the additional scaffold-tree filter used in the pretrained-model experiments. None of the 64 evaluation compositions occurs in either selected subset or either full processed corpus.

Optimization. Each model receives 30,000 AdamW updates at learning rate 10^{-4} and batch size 32, for 960,000 training presentations. Each batch contains molecules with the same atom count. We sample atom-count groups in proportion to their sizes, so each training row has the same marginal

Table 9: Graph validity of the separately adapted conditional EDM, before geometry optimization. Each run has 768 attempts. For context, the harmonic and force-updated harmonic models in Table 15 reach 52.73% and 53.45% at 128 calls, but use a different pretrained backbone.

Sampler	Run 1	Run 2	Pooled rate
EDM, 128 calls	103	100	13.22%
EDM, 1,001 calls	93	106	12.96%

Table 10: Graph validity on the 64-composition set for models trained from scratch on the same 2.38-million-parameter EGNN. Each seed has 2,048 generation attempts at 128 network forward passes per output. Counts are graph-valid outputs; the final column pools both seeds.

Method	Valid, seed 1	Valid, seed 2	Pooled rate (%)
Gaussian FM	182	218	9.77
Harmonic FM	223	232	11.11
EDM	301	216	12.62
GAGA	321	249	13.92

sampling probability. We clip the gradient norm at one. Generation uses an exponential moving average of weights at the final update, with decay $\min(0.999, (1 + s)/(10 + s))$ at optimizer step s . Internal validation at 5,000, 15,000, and 30,000 updates monitors learning; it is not used to choose a checkpoint.

Flow-matching models. Gaussian and harmonic FM share the linear interpolation, atom matching, and rotation alignment. Both predict velocity in one EGNN pass and use 64 midpoint steps, giving 128 network evaluations. Neither uses self-conditioning. The harmonic distribution is the same as in the main model, but its factorized edge weights permit a different exact tree sampler. We draw the $N - 2$ entries of a Prüfer sequence independently with probabilities $u_i / \sum_j u_j$ and decode the sequence into a tree. This samples the same tree law used to place atoms.

Diffusion models. EDM predicts Gaussian coordinate noise under its original variance-preserving schedule. Position-conditional GAGA (Qu et al., 2026) truncates this schedule at 650 of 1,000 steps, using the published GEOM setting. Its initial Gaussian has variance $\alpha_t^2 v + \sigma_t^2$, where α_t and σ_t are the signal and noise coefficients. The training-only estimate $v = 3.0232257866 \text{ \AA}^2$ is the mean squared centered coordinate per independent coordinate. We transfer the truncation setting from GEOM without optimizing it on OMol25 evaluation results.

Generation and compute. Every model generates 32 raw structures per evaluation composition at 128 network evaluations. We also evaluate the complete EDM and GAGA reverse schedules at 1,001 and 651 calls, including their final observation steps. This experiment uses no physical energy labels, bond supervision, energy-based selection, or geometry optimization. Scheduler metadata record the A100, H200, and A100 MIG allocations. Equal training presentations and network-call counts do not imply equal wall-clock time.

Table 11: Diffusion results at the shared 128-call budget and the full reverse schedules. Each independent initialization generates 2,048 outputs. Counts give graph-valid outputs; the final rate pools both initializations.

Method	Evaluations	Valid, seed 1	Valid, seed 2	Rate (%)
EDM	128	301	216	12.62
EDM	1,001	308	206	12.55
GAGA	128	321	249	13.92
GAGA	651	273	267	13.18

K.1 GEOMETRY FEEDBACK AND PHYSICAL TRANSFER ON THE EGNN

We test geometry feedback on the independent EGNN using the same 20,000 training rows and two initializations. The first pass predicts provisional coordinates; the second uses their pairwise distances in an existing invariant edge channel. The provisional geometry is detached, and both passes are supervised. Thus this experiment uses 15,000 updates at batch size 32 and two forward passes per example, matching the 960,000 training forward passes of the original one-pass models. The total parameter count is 2,381,566, including 1,001 frozen noise-schedule entries. Optimizer-update counts and wall time are not matched.

An alternative replaces distances with edge-inclusion probabilities from a regularized auxiliary tree distribution. On 32 internal validation compositions, we select distance or tree feedback by pooled graph validity, with one choice for both seeds. GAGA separately selects its maximum sampling timestep from 350, 500, and 650. Validation selects distance feedback and timestep 650. We then evaluate 32 previously ungenerated compositions, eight in each of the four size ranges, with 32 draws per composition and seed. Each output uses 128 network calls, without physical refinement.

Table 12: Raw graph validity on the feedback-confirmation panel. Each model has 1,024 attempts per initialization. All tested feedback choices are shown.

Model	Valid, seed 1	Valid, seed 2	Pooled (%)
Harmonic FM, one pass	107	113	10.74
Harmonic FM, distance feedback	155	165	15.63
Harmonic FM, tree feedback	114	84	9.67
GAGA, selected timestep 650	151	131	13.77

Distance feedback exceeds GAGA by 1.86 points, with a composition-bootstrap 95% interval of $[-0.34, 4.44]$. The seed-specific differences are +0.39 and +3.32 points. Tree feedback reduces validity relative to distance feedback by 5.96 points ($[-8.11, -3.96]$), so it is not part of the proposed method.

Transferring the fixed physical-update recipe. Each selected parent generates 512 candidate anchors on eight training compositions. We apply the same local force-target procedure, then fit physical and replay students. Each FM student uses 1,000 two-pass updates; each GAGA student uses 2,000 one-pass updates. The learning rate is 10^{-4} and the parameter-difference coefficient is one. This is a follow-up on the same 32-composition panel. It tests this fixed recipe, without selecting its strength on validation.

Table 13: Fixed-strength physical adaptation on the EGNN. Joint yield requires graph validity and GFN2 force RMS at most 5 eV/Å, with all 2,048 attempts per model in the denominator.

Model	Graph validity (%)	Joint yield (%)
FM with distance feedback	15.63	4.79
FM + paired physical update	2.00	0.78
GAGA	13.77	5.86
GAGA + paired physical update	4.79	1.90

The full-strength update lowers both graph validity and joint quality for both families. The within-FM joint-yield change is -4.00 points ($[-6.49, -2.00]$); the within-GAGA change is -3.96 points ($[-6.25, -1.95]$). eSEN gives the same directions. These results show that the successful primary-backbone recipe cannot be transferred unchanged to this EGNN. GAGA’s predefined noise schedule stays fixed in this comparison. Appendix N evaluates a separately trained correction head with broader supervision from each parent’s own trajectory states.

L PHYSICAL FEEDBACK THROUGH PAIRED TRAINING TARGETS

We next fine-tune the generator using nearby structures constructed from physical forces. A frozen harmonic generator produces training structures, called *anchors* A , on compositions excluded from evaluation. For each graph-valid anchor, we draw nearby coordinates from $q_A = \mathcal{N}_{\mathcal{H}_N}(A, \sigma_A^2 I)$. The width σ_A controls the size of the perturbations within the centered coordinate space.

The teacher is a target-construction procedure based on the learned eSEN energy model. We average energies and forces over mirror-related coordinates, giving E_+ and F_+ (Appendix T). Each perturbed structure is shifted by the same anchor force: $Y = X + \delta_A$, where $\delta_A = \sigma_A^2 F_+(A)/(k_B T_{\text{phys}})$ and $T_{\text{phys}} = 300$ K. Here k_B is Boltzmann’s constant. This *force escort* is motivated by a local approximation: if energy is linear near A , weighting the Gaussian by a Boltzmann factor shifts its mean by exactly δ_A . We bound the displacement by choosing σ_A before drawing candidates. We retain only candidates that pass the geometric checks and preserve the anchor’s inferred labeled graph, then sample uniformly from them. Appendix T gives the bounds, restriction, and derivation.

We fine-tune two copies of the base weights θ_{base} . The physical copy learns from the force-shifted targets, producing θ_{physical} . The replay copy learns from the original anchors, producing θ_{replay} . Both also train on the same OMol25 references and follow the same schedule. Replay provides a control for further training on generated structures without the physical displacement. We form the final weights as

$$\theta_{\text{final}} = \theta_{\text{base}} + (\theta_{\text{physical}} - \theta_{\text{replay}}). \quad (8)$$

This parameter arithmetic (Ilharco et al., 2023) is applied elementwise with coefficient one. It changes the update direction rather than simply retaining the directly fine-tuned physical model. The result is one network, not an ensemble. Section L.1 and Appendix T compare it with direct fine-tuning and a norm-matched control.

Distributional extensions. We also study *complete-work* weights, which account for the energy change, the local Gaussian density, and any volume change introduced by the coordinate map. Appendices S–U derive these weights and test recovery of specified local distributions. The paired-update generator uses the simpler, uniformly sampled force-shifted targets. Recovering a particular local distribution and reducing generated energy are different objectives.

Connectivity and geometry of individual outputs. For the paired-update variant, the first four graph-valid outputs with distinct connectivities from the first eligible CHON composition in run 1 contain two, two, two, and five independent cycles. These archived examples were selected without energy ranking or geometry optimization. One passes the additional geometry screen in Appendix C; the others fail angle, planarity, or internal-distance checks. The per-example checks and original coordinates are included with the released evidence. Figure 1 shows a separate raw output that passes the stated geometric checks, together with its inferred two-dimensional connectivity.

L.1 WHY SUBTRACT THE REPLAY UPDATE

Does replay subtraction help because it changes the update direction, or simply because it reduces the update size? We compare the paired update with direct physical fine-tuning and a direct update rescaled to the same parameter norm. The rescaled control is $\theta_{\text{control}} = \theta_{\text{base}} + \alpha(\theta_{\text{physical}} - \theta_{\text{base}})$, where α is the ratio of paired to direct update norms. Each norm is the square root of the summed squared changes in model parameters; buffers are excluded. The coefficients are 0.2536 and 0.2564 in the two runs, determined from the trained weights before generating evaluation samples.

The rescaled direct update largely preserves graph validity but increases energy relative to the base model. The paired update lowers energy instead. Paired minus same-norm direct gives -0.03232 eV/atom in energy $([-0.03981, -0.02485])$ and -0.4170 eV/Å in force RMS $([-0.5529, -0.2859])$. Both runs favor the paired direction, and comparisons using all outputs have the same signs. The graph-validity difference is $+0.39$ percentage points, with interval $[-1.95, 2.73]$. These results support a benefit from the replay-subtracted direction, rather than update size alone.

Table 14: Update controls on 10 compositions, with 1,280 generation attempts per model. Energy changes are measured with eSEN relative to the base model, using paired outputs that are graph-valid under both models.

Model	Graph validity (%)	Energy change (eV/atom)
Base	62.11	0
Direct physical update	54.77	+0.11080
Same-norm direct update	61.64	+0.01893
Paired update	62.03	−0.01554

L.2 NEW COMPOSITIONS AND INDEPENDENT PHYSICAL EVALUATION

We evaluate four frozen models on 24 new compositions: Gaussian, harmonic, harmonic with the force-only update, and harmonic with the complete-work update. The force-only model was selected as the primary physical model before this test. Eight compositions fall in each of the 17–20, 21–24, and 25–28 atom ranges; none appears in either processed corpus. There are 6,144 raw outputs in total. Appendix F describes the fixed metadata-based selection.

The harmonic source improves graph validity over Gaussian by 8.20 percentage points (95% CI [5.08, 11.33]), with gains in both runs. Adding the force update changes harmonic validity by +0.72 points, with interval [−0.98, 2.34]. Thus the measured validity rate remains similar after physical fine-tuning. Every accepted output has a distinct inferred connectivity within its composition in these draws.

Energy and force scores. We evaluate every output at its saved coordinates using eSEN and gas-phase GFN2-xTB single-point energies and gradients (Bannwarth et al., 2019). GFN2 is used only for evaluation, independently of the eSEN teacher. Four of the 6,144 generated structures have failed GFN2 calculations; all four are graph-invalid. Calculations succeed for every graph-valid output, and reference inversion checks pass. eSEN uses the mirror averages described in Section 4. Force RMS is the root mean squared magnitude of the atomic force vectors.

The force update lowers energy and force RMS among paired, jointly valid outputs under both potentials; every composition contributes pairs in both runs. The directions also hold across all eSEN outputs and all paired successful GFN2 calculations, regardless of graph validity. Without physical updates, harmonic also lowers aggregate energy relative to Gaussian, predominantly in one run.

Complete-work and force-only updates have similar measured scores. On paired, jointly valid outputs, the complete-work-minus-force energy difference is −0.00413 eV/atom under eSEN (interval [−0.00903, 0.00078]) and −0.00363 under GFN2 ([−0.00792, 0.00075]). Force and graph-validity intervals also include zero. We therefore use the cheaper force-only teacher as the primary method; these comparisons do not resolve an additional benefit from work weighting.

L.3 SOURCE AND PHYSICAL ADAPTATION

We apply paired physical training to both source families, using generated anchors from the same eight training compositions and matched maximum proposal and training budgets. Table 17 compares all four models on 24 compositions. Appendix W.2 records preparation costs and evaluation reuse.

The paired update improves GFN2 joint yield for both sources: by 4.95 percentage points for Gaussian (composition-bootstrap 95% interval [2.86, 7.10]) and 3.06 points for harmonic ([1.04, 5.14]). Both runs improve for both sources. On paired, jointly valid outputs, the Gaussian update also lowers energy by 0.01465 eV/atom and force RMS by 0.4618 eV/Å.

After physical adaptation, harmonic exceeds Gaussian by 6.84 points in pooled graph validity ([4.23, 9.51]) and 6.97 points in joint yield ([3.26, 10.61]). The validity difference is positive in both runs, but the joint-yield differences are +16.28 and −2.34 points. Thus the pooled advantage coexists with substantial variation between fine-tuning runs. eSEN also shows the within-source benefits of physical adaptation and the pooled harmonic-source advantage. Figure 13 in Appendix W plots the results across force thresholds.

Table 15: Graph validity on 24 new compositions, before geometry optimization. Each run has 768 attempts per model, using 128 network forward passes per attempt. The middle columns count valid outputs; the last column pools both runs.

Model	Run 1	Run 2	Pooled rate
Gaussian reference	332	352	44.53%
LatentSpring harmonic	435	375	52.73%
With force update	439	382	53.45%
With complete-work update	426	383	52.67%

Table 16: Effect of the primary force update relative to the frozen harmonic model. The comparison uses paired outputs that are graph-valid under both models. Negative values indicate lower energy or residual force. Both fine-tuning runs improve under both potentials; brackets show paired 95% confidence intervals.

Potential	Energy change (eV/atom)	Force-RMS change (eV/Å)
eSEN	-0.01308 [-0.01805, -0.00817]	-0.34793 [-0.41498, -0.28478]
GFN2	-0.01168 [-0.01616, -0.00724]	-0.35620 [-0.42983, -0.28714]

M LEARNING A PHYSICAL CORRECTION ALONG THE FLOW

The correction network learns physical changes at provisional endpoints while keeping every parent parameter fixed. Its regression target is separate from the backbone’s flow-matching objective.

States and targets. The parent generates two trajectories for each of 128 training compositions, with 32 compositions in each of the 8–16, 17–24, 25–32, and 33–40 atom ranges. We record the second velocity evaluation at midpoint steps 23 and 29, corresponding to flow times 0.734375 and 0.921875. These 256 trajectories yield 512 states. At a recorded state, the parent velocity defines the provisional endpoint $H_t = X_t + (1 - t)v_{\text{base}}(X_t, t; c)$. This is an extrapolation from the current prediction after both parent self-conditioning passes; later changes in velocity can change the final generated coordinates.

We evaluate the mirror-averaged eSEN force at H_t , subtract its mean across atoms, and construct the force shift defined in Equation 5. The Gaussian width is at most 0.03 Å and the molecular displacement norm is at most 0.1 Å. The regression target is $\delta(H_t)/(1 - t)$, a velocity correction whose linear endpoint extrapolation has displacement $\delta(H_t)$. All recorded states are retained, including states whose provisional geometry fails the chemical-graph check. Compositions are separated by metadata hashes: 96 train the head, and 32 provide internal validation, with four recorded states per composition. These are subsets of the parent’s training compositions, separate from generator confirmation conditions.

Equivariant pair correction. The small model has 8,178 parameters. For each unordered atom pair, it computes symmetric scalar features from atomic embeddings, the two pair distances in X_t and H_t , their directional alignment, time, atom count, and covalent radii. A two-hidden-layer network, each layer containing 64 units with SiLU activations, predicts two coefficients α_{ij} and β_{ij} . A hyperbolic tangent bounds both coefficients between -1 and 1.

Write $r_{ij} = X_{t,i} - X_{t,j}$ and $\hat{r}_{ij} = H_{t,i} - H_{t,j}$. We normalize these vectors as $\bar{r}_{ij} = r_{ij}/\sqrt{\|r_{ij}\|^2 + \ell_{ij}^2}$ and likewise for $\bar{\hat{r}}_{ij}$, using the covalent-radius sum ℓ_{ij} . The contact weight $w_{ij} = \text{sigmoid}((1.25 - \|\hat{r}_{ij}\|/\ell_{ij})/0.2)$ decreases as the predicted atoms move apart. It is a geometric weight, not a supplied bond label. Let $d_i = \sum_{j \neq i} w_{ij}$ and $d_{\max} = \max(1, \max_i d_i)$. The correction for atom i is

$$A_{\phi,i} = v_{\text{cap}} t^2 \sum_{j \neq i} \frac{w_{ij}}{2d_{\max}} \left(\alpha_{ij} \bar{r}_{ij} + \beta_{ij} \bar{\hat{r}}_{ij} \right), \quad v_{\text{cap}} = 2 \text{ Å per unit flow time.} \quad (9)$$

The symmetric coefficients and opposite pair vectors give zero total correction. Rotating the coordinates rotates the correction, and relabeling atoms relabels its outputs. These pair constructions

Table 17: Source and physical-update comparison under GFN2-xTB, with 1,536 raw attempts per model. Median force RMS uses each model’s own graph-valid outputs with successful calculations. Joint yield uses all attempts and requires both graph validity and force RMS at most 5 eV/Å.

Model	Graph validity (%)	Median force (eV/Å)	Joint yield (%)
Gaussian source	44.53	3.404	35.68
Gaussian + paired update	46.61	3.222	40.63
Harmonic source	52.73	3.287	44.53
Harmonic + paired update	53.45	3.037	47.59

follow standard equivariant-network practice (Satorras et al., 2021). The triangle inequality gives $\|A_{\phi,i}\| \leq v_{\text{cap}} t^2$ for every atom. This bounds the instantaneous velocity addition, not the final displacement after interaction with the parent flow. The final layer starts at zero, reproducing the parent field before training.

We fit the correction with mean squared velocity error, AdamW at learning rate 3×10^{-4} , batch size one, and gradient clipping at one. Two seeds use 20,000 updates each. Only the head is optimized. We compare a 37,540-parameter pair network with wider layers and a 37,586-parameter alternative with two rounds of neighborhood message passing and invariant angular summaries. All receive the same training-state sequence and target vectors within a seed.

On the internally withheld compositions, the contextual model has about 15% lower target-prediction error than the small pair model and about 11% lower error than the nearly parameter-matched wide model at 20,000 updates. The confirmation experiment below tests whether this predictive difference improves actual generation. Physics-based residual correction of frozen flow fields has also been studied for video generation (Wang et al., 2026a); here supervision is provided by molecular forces at predicted atomic geometries.

Sampling and preparation cost. Generation integrates $v_{\text{base}} + A_{\phi}$ with the same 32 midpoint steps. Each attempted structure uses 128 backbone evaluations and 64 head evaluations, with no terminal noise, energy call, output filtering, or geometry optimization. The parent control carries a zero-output head so its sampling operations match. The training-state collection uses 256 parent trajectories. Mirror-paired force labels for the 512 recorded endpoints require 1,024 eSEN queries. The study also queries eight translated Gaussian particles per state for a complete-work control, bringing the actual preparation expenditure to 9,216 queries. The force-target heads do not use those particle energies. The local-work teacher has median particle effective sample size 1.37 out of eight; its earlier 2,000-update student comparison did not establish an additional generation gain.

N PHYSICAL CORRECTION ON A SHARED BACKBONE

We test whether the learned physical correction transfers to the independent EGNN, and give GAGA the same correction architecture and physical-supervision budget. Both parents have 2,381,566 parameters and use the same 20,000 OMol25 training rows. The FM parent uses distance self-conditioning. Its 15,000 two-pass updates match the 960,000 training-example forward passes of GAGA’s 30,000 one-pass updates, as described in Appendix K.1. Two independent parent initializations each receive a separately fitted head.

A common predicted endpoint change. FM predicts a velocity, whereas GAGA predicts noise. We express the head’s action through the change it induces in the parent’s provisional final geometry. Write p for the fraction of the sampling schedule already completed. For FM, p is the flow time; for GAGA, $p = 1 - t/t_{\text{max}}$, where t is diffusion noise time and $t_{\text{max}} = 0.65$ is the selected truncation. These are normalized schedule positions, not a shared physical time.

The head A_{ϕ} takes the current coordinates X , provisional final geometry H , atom identities, and progress p . It predicts the same bounded vector correction as in Appendix M. A strength η scales its implied endpoint displacement to $\eta(1 - p)A_{\phi}$. The corresponding updates to the two parent outputs

are

$$H = X + (1 - p)v, \quad v^{\text{corr}} = v + \eta A_\phi, \quad (10)$$

$$H = \frac{X - \sigma_t \epsilon}{\alpha_t}, \quad \epsilon^{\text{corr}} = \epsilon - \eta \frac{\alpha_t(1 - p)}{\sigma_t} A_\phi. \quad (11)$$

Here v is the FM velocity, ϵ is GAGA’s predicted noise, and α_t, σ_t are the signal and noise scales of its fixed schedule. Substitution shows that both equations change the provisional endpoint by $\eta(1 - p)A_\phi$. The native GAGA transitions and final observation noise are preserved. Every parent tensor, including its noise schedule, remains frozen. With the common 16-element embedding vocabulary, each head has 7,106 parameters; its hidden pair network is the same as the small FlowMol correction.

Physical supervision and selection. Each parent generates two trajectories for each of the same 128 TRAIN compositions, balanced across the four atom-count ranges in Appendix M. Two late field evaluations per trajectory give 512 provisional endpoints. FM records progress 0.734375 and 0.921875; GAGA records the nearest reverse-sampling evaluations, at progress 1 – 174/650 and 1 – 51/650. All states are retained. Mirror-paired eSEN queries provide centered forces, and the same bounded displacement rule gives the regression target $\delta(H)/(1 - p)$ for A_ϕ . No target requires capping by the per-atom velocity bound. Each head uses 20,000 AdamW updates on 96 TRAIN compositions; the other 32 measure target-prediction transfer. The four fits reduce this internal prediction error by 29.7–32.1% relative to zero correction.

Eight separate validation compositions select one strength per algorithm across both fits from $\{0, 1/4, 1/2, 1\}$. Selection maximizes graph-valid yield with GFN2 force RMS at most 5 eV/Å, including every attempt in the denominator; ties choose the smaller strength. Both algorithms select $\eta = 1$. Only after this choice is audited and frozen do we generate the new test panel: 16 compositions, eight each with 17–28 and 29–40 atoms, with 16 outputs per composition and fit. None occurs in the prepared training or validation corpus, or in earlier generator evaluation panels.

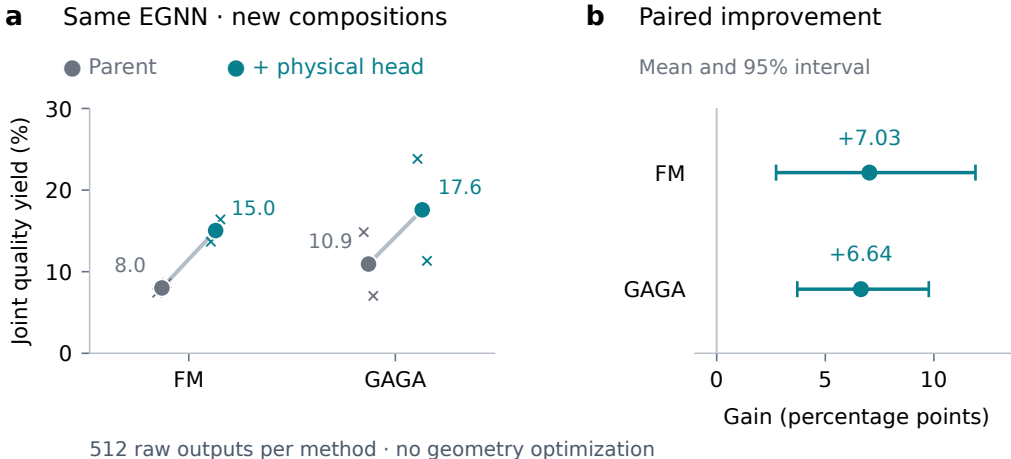


Figure 7: Physical correction transfers to independently trained EGNN generators. (a) Joint yield requires an inferred valid graph and GFN2 force RMS at most 5 eV/Å. Large circles show pooled rates; small crosses show the two fits. (b) Within-parent improvements with paired composition-bootstrap 95% intervals. Both algorithms receive the same TRAIN compositions, force-query budget, head architecture, and fitting schedule. All generated coordinates are unoptimized.

The primary FM comparison improves joint yield by 7.03 percentage points (95% interval [2.73, 11.91]), with gains of 6.25 and 7.81 points in the two fits. GAGA also benefits, gaining 6.64 points ([3.71, 9.77]), with both fits improving. Thus the learned correction transfers beyond the pretrained FlowMol backbone and works with both velocity and noise prediction.

After identical physical supervision, FM minus GAGA is -2.54 points ($[-6.64, 1.76]$). The difference between their within-parent improvements is $+0.39$ points ($[-4.30, 5.47]$). The experiment supports

Table 18: Fresh shared-EGNN confirmation. Each row contains 512 attempts on the same 16 compositions. The last two columns give joint yield in each independent parent/head fit. Every GFN2 single-point calculation completed successfully.

Model	Graph (%)	Joint (%)	Fit 1 (%)	Fit 2 (%)
FM, distance feedback	17.58	8.01	7.42	8.59
FM + physical head	20.31	15.04	13.67	16.41
GAGA	18.75	10.94	14.84	7.03
GAGA + physical head	20.31	17.58	23.83	11.33

physical-head transfer to both generators, while their ordering remains uncertain. Intervals resample compositions and condition on the two fitted parents and heads; validation strengths are fixed before test generation.

Cost and verification. The study uses 1,024 TRAIN trajectories, 4,096 eSEN teacher queries, and 80,000 head updates in total. Validation and test each contribute 2,048 raw outputs and the same number of fixed-coordinate GFN2 calculations. Each output uses 128 backbone calls; FM uses 64 additional head calls and GAGA uses 128. The zero-strength controls execute the same head operations as their corrected counterparts. Supervision and head-update budgets match; total FLOPs and wall time are not equated. Generation uses no physical queries, geometry optimization, or energy-based selection. Zero heads and passive trajectory observers reproduce the native samplers exactly. Auditing reconstructs the targets, checks fitted head errors, re-evaluates all raw graphs, and independently parses xTB gradients.

N.1 CORRECTION STRENGTH AND THE QUALITY TRADEOFF

The validation study above selects the largest allowed strength for both generators. We extend the common grid to $\{0, 1/4, 1/2, 1, 2, 4\}$, reusing the same eight validation compositions and random seeds. Only strengths two and four require new samples. Parents, heads, force labels, and fitting budgets are unchanged. The applied correction bound scales with the strength η ; the bound stated for $\eta = 1$ is not retained at larger strengths.

We compare two validation choices. A conservative rule permits at most a one-point decrease in pooled graph validity and a three-point decrease in each fit, relative to strength one. It retains strength one for both generators. Selecting solely by joint yield chooses strength four for both, with validation joint yields of 31.64% for FM and 28.91% for GAGA. This second objective is declared after the conservative screen, before any new test generation. Both choices are then evaluated on the same 32 previously ungenerated compositions, 16 each with 17–28 and 29–40 atoms, with 16 outputs per composition and fit.

Table 19: Strength confirmation on the shared EGNN, with 1,024 unoptimized outputs per row. Force is the median RMS over each model’s graph-valid, successfully scored outputs. Sampling time is measured per output within the same GPU allocation for each fit, excluding checkpoint loading and physical assessment.

Model	Graph (%)	Joint (%)	Force (eV/Å)	Time (s)
FM, $\eta = 1$	20.12	15.14	3.833	0.123
FM, $\eta = 4$	22.17	20.80	3.159	0.122
GAGA, $\eta = 1$	20.80	16.50	3.805	0.143
GAGA, $\eta = 4$	21.00	20.02	3.158	0.143

Stronger correction improves FM joint yield by 5.66 percentage points (95% composition-bootstrap interval $[2.73, 9.38]$), with gains of 7.23 and 4.10 points in the two fits. The GAGA change is +3.52 points ($[0.00, 7.52]$). At strength four, FM minus GAGA is +0.78 points in joint yield ($[-4.00, 6.54]$) and +1.17 points in graph validity ($[-2.93, 5.86]$). The fit-specific joint differences are -1.37 and $+2.93$ points, so the pooled ordering is not stable across fitted models. The recorded FM sampling time is 14.6% lower, with 64 head calls per output versus GAGA’s 128; both use 128 backbone

calls. These are measured sampling times, not equal training cost or a hardware-independent speed guarantee.

Across all outputs, stronger correction lowers FM’s mean GFN2 energy by 0.0622 eV/atom, with signed-change interval $[-0.0769, -0.0479]$. At strength four, FM’s mean energy remains 0.0527 eV/atom above GAGA’s ($[0.0273, 0.0786]$). These composition-paired readouts include invalid outputs and different inferred connectivities; they are not equilibrium free-energy differences. All graph-valid outputs within each 16-draw composition/fit cell have distinct canonical connectivities when stereochemistry is omitted.

The remaining errors are predominantly structural. At strength four, 533 of 1,024 FM outputs and 579 GAGA outputs fail the geometric connectedness test; only two FM outputs and no GAGA outputs overlap at the prescribed radius threshold, with both overlaps among the disconnected cases. Of the graph-valid outputs, 213/227 for FM and 205/215 for GAGA pass the force threshold. The extension adds 1,024 validation outputs and 4,096 confirmation outputs, each receiving a GFN2 attempt, with no new fitting or eSEN calls. One GAGA strength-one GFN2 calculation fails and remains in the denominator; the other 4,095 confirmation calculations succeed.

N.2 FROZEN-METHOD TRAINING REPLICATION

We freeze all architectures, checkpoint schedules, physical strength four, and hydrogen-flow settings before the new fits. Each new FM parent receives 15,000 two-pass updates of batch size 32; GAGA receives 30,000 one-pass updates. Initialization tensors and the batch-schedule prefix match within each pair. This matches backbone training-example forward passes, while optimizer steps, data presentations, and measured runtime differ. All newly trained parents, heads, and H flows use their prescribed final checkpoints.

The new metadata pool excludes every composition in the previous pool. Exhaustive qualification leaves only 20 previously unused compositions with 17–28 atoms. Before any generation, we therefore fix these 20 and the first 44 eligible compositions with 29–40 atoms under the previously specified metadata ranking. The planned 32-per-bin selection and its failed qualification are retained. The final 64 compositions occur in neither processed generator corpus nor the prepared training/validation rows. Reference geometry and overlap checks use no generated outcomes or energy ranks.

For each parent, the strength-zero and strength-four trajectories share source draws and sampler randomness. The zero-strength control evaluates the head but multiplies its output by zero; its recorded call budget therefore includes these unused head calls. Each corrected output also receives the unchanged, fixed radial, and learned hydrogen readouts. Original graph-valid outputs are preserved exactly by the H readout. Identical coordinates reuse their original physical calculation, and all failed calculations remain in graph/force denominators.

The primary independent-training analysis uses the three new pairs. The two earlier pairs participated in method development and are included in the separate five-pair summary. Composition intervals resample the 64 compositions while retaining all fitted models. Crossed intervals resample both fit indices and composition indices; only three new fits, or five total fits, are available to estimate training variation. We additionally give equal-weight averages of the two size bins, so the 20:44 composition mix is explicit.

Table 20: Joint yield in every fit (percent). Fits 1–2 reuse earlier weights; fits 3–5 are newly trained.

Parent	Module	1	2	3	4	5
FM	None	9.96	7.52	3.61	6.25	8.69
FM	Physical	28.22	26.37	11.72	24.71	29.00
FM	Physical + radial	30.18	27.83	12.70	27.15	32.42
FM	Physical + H flow	30.66	28.03	12.89	27.34	32.71
GAGA	None	12.30	6.64	11.52	8.59	4.69
GAGA	Physical	27.34	24.02	28.61	27.34	19.53
GAGA	Physical + radial	27.93	24.22	29.20	27.83	19.73
GAGA	Physical + H flow	27.93	24.22	29.00	27.83	19.73

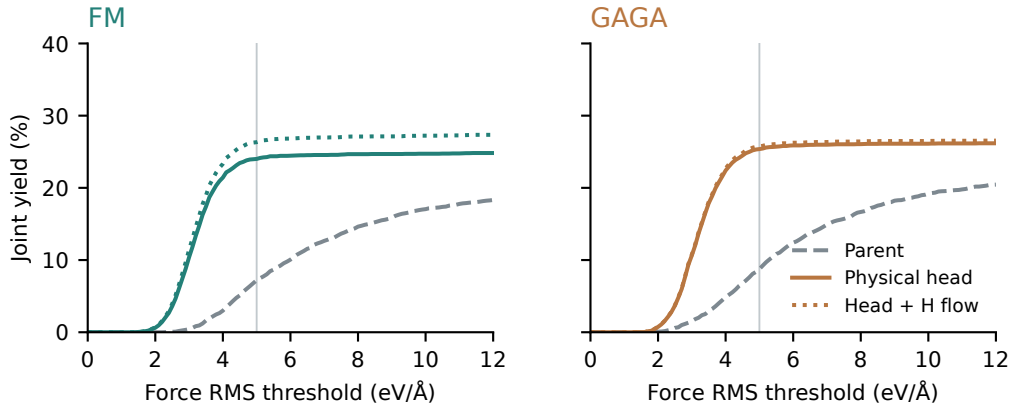


Figure 8: Joint yield as the force threshold varies, averaged over all five fits. The vertical reference marks the prespecified 5 eV/Å endpoint. These curves are descriptive; no threshold is selected from them.

fm physical minus parent: pooled change +16.80 points, crossed interval [11.50, 21.93]; equal-stratum change +19.34 points, crossed interval [13.85, 24.69].

gaga physical minus parent: pooled change +16.62 points, crossed interval [12.95, 20.47]; equal-stratum change +18.46 points, crossed interval [14.32, 22.72].

fm minus gaga hydrogen: pooled change +0.59 points, crossed interval [-8.59, 8.59]; equal-stratum change +0.95 points, crossed interval [-8.93, 9.63].

The study adds 20,480 evaluation parent trajectories, 20,480 derived readout outputs, and 22,688 GFN2 attempts. New physical teachers use 1,536 TRAIN trajectories and 6,144 eSEN calls. The three new pairs add 135,000 parent updates, 120,000 head updates, and 30,000 hydrogen-model updates. Reused parent and readout calculations are not counted as new oracle calls.

Six GFN2 single-point calculations initially fail to converge electronically, all on graph-invalid geometries. A uniform numerical-completion rule increases the SCC iteration limit from 250 to 1,000 while preserving coordinates, charge, spin, electronic temperature, Hamiltonian, and convergence accuracy. All six then converge at this first recovery stage. The original failures remain archived; completing their energies changes no graph or joint count. The all-output energy differences use these completed scores. Among 1,384 paired FM outputs that are graph-valid under both hydrogen readouts, 111 are newly repaired under both. On those 111 descriptive pairs, the learned readout lowers mean energy by 83.42 meV/atom relative to the radial rule.

N.3 CROSS-GENERATOR WEIGHT TRANSFER

The transfer protocol is fixed before any cross-head outputs are generated and before the three new fit pairs are evaluated. Earlier target-parent outcomes already exist for fits 1–2. Fits 3–5 therefore provide the primary independent replication. Each target uses the source family’s final 20,000-update head without changing any tensor, vocabulary, bound, or calibration strength. The velocity/noise conversions in Equation 6 express its action in the target sampler. No backbone representation is supplied to the head; the inputs are coordinates, predicted endpoints, atom identities, and normalized progress.

Table 21: Joint yield of transferred heads in every fit (percent), without hydrogen readout.

Target	Head trained on	1	2	3	4	5
FM	GAGA	29.20	28.81	13.96	26.17	29.79
GAGA	FM	28.42	23.54	27.54	26.95	19.92

Relative to its own head, GAGA with the FM-trained head changes joint yield by -0.36 points in the three new fits (composition interval $[-1.66, 0.94]$; crossed interval $[-2.15, 1.30]$). For FM, the GAGA-trained head changes joint yield by $+1.50$ points ($[0.33, 2.64]$; crossed interval $[0.00, 3.12]$). The robust conclusion is improvement over the uncorrected target in both transfer directions. These results do not establish that transferring a head is always better than fitting one on the target’s own trajectories.

This study adds 10,240 generated trajectories and 10,240 successful GFN2 single-point calculations, with no further head updates or force-teacher queries. All five fit pairs and both directions are retained. A resumed metadata-writing failure reuses its checksum-verified generated coordinates; those trajectories are counted once.

N.4 A CONDITIONAL FLOW FOR HYDROGEN COORDINATES

A connected heavy-atom skeleton can still fail molecular validation if a hydrogen lies too far from every heavy atom. In 512 cached training outputs per generator, FM produces 346 connected heavy skeletons and GAGA 314, but hydrogen-only fragmentation affects 60 FM outputs and 18 GAGA outputs. This motivates a conditional coordinate model that holds the heavy-atom geometry fixed and learns how the hydrogens should be arranged around it. Hydrogen decoration is also supported by existing generators such as Quetzal (Cheng et al., 2025); here we test it as a selective readout of a generated structure, with explicit geometric and physical controls.

Inputs, targets, and training. The model receives all current coordinates, atomic numbers, and a completion time $s \in [0, 1]$. Its output is a three-dimensional velocity for each hydrogen. For a training structure, we perturb hydrogen coordinates with Gaussian noise of standard deviation $0.4 \text{ \AA}$, keeping heavy atoms fixed. Half of the examples perturb every hydrogen; the other half perturb each hydrogen independently with probability one half. We pair perturbed and reference hydrogen positions by minimum squared-distance assignment, so the target does not depend on arbitrary labels of identical hydrogens. If Y_0 denotes the perturbed hydrogen coordinates and Y_1 their assigned reference positions, the training path and loss are

$$Y_s = (1 - s)Y_0 + sY_1, \quad \mathcal{L}_H = \mathbb{E} \left[\frac{1}{3N_H} \|u_\psi(X_{\text{heavy}}, Y_s, s; z) - (Y_1 - Y_0)\|_F^2 \right]. \quad (12)$$

Here X_{heavy} contains the fixed heavy-atom coordinates, z lists all atomic numbers, N_H is the hydrogen count, and u_ψ is the predicted hydrogen velocity. A common translation centers the full input; it does not change relative coordinates. Training uses coordinates and atom identities, without bond labels, energies, or forces.

We use a four-layer, width-64 EGNN with 153,790 total parameters, including 1,001 frozen schedule entries. Two independently initialized models each train for 10,000 updates of batch size 32 on the same 20,000 OMol25 structures used by the shared-backbone generators. The learning rate is $3 \cdot 10^{-4}$ and the exponential moving average decay is 0.999. We use the final averaged checkpoint. These fits add 640,000 training-example forward passes in total; the same two hydrogen models serve both FM and GAGA.

Generation and controls. The parent generators use their frozen strength-four physical corrections. A hydrogen activates the readout when its distance from every heavy atom exceeds 1.25 times their summed covalent radii. With no such hydrogen, the entire output is unchanged. Otherwise we integrate the conditional flow from $s = 0$ to 1, allowing all hydrogens in that molecule to move. Four midpoint steps require eight small-network evaluations per activated output. Each predicted hydrogen velocity is capped at $2 \text{ \AA}$ per unit completion time. Heavy atoms remain fixed during integration, and a final common translation restores the input centroid. Thus their relative geometry is preserved. No force or energy evaluator is called during this readout.

The fixed radial control moves each detached hydrogen toward its nearest heavy atom until their separation equals the sum of covalent radii. It leaves all other relative coordinates fixed. A second control applies the learned flow only to the hydrogens that triggered the readout, instead of moving all hydrogens together. Both controls start from exactly the same parent outputs. These are coordinate readouts, distinct from both the unchanged parent and force-field or quantum geometry optimization.

Selection and independent physical test. Sixteen development compositions compare the radial rule with the two learned modes, each starting at time zero or one half. Moving all hydrogens from time zero gives the highest FM joint yield, 96/256 versus the radial rule’s 95/256, but does not meet the requirement of improvement in both fits. Its lower energy motivates a separate physical hypothesis: relative to the radial rule, the learned readout lowers mean energy per atom while retaining pooled graph and joint yield. This hypothesis and the frozen model are specified before generation on 30 new compositions, 15 each with 17–28 and 29–40 atoms. Each fit draws 16 samples per composition, giving 960 attempts per setting. Composition bootstrap intervals retain both fitted models in every resample.

Table 22: Hydrogen readouts on 30 fresh compositions. All rows use the same strength-four physical heads. Joint yield additionally requires GFN2 force RMS at most 5 eV/Å. Energy is the all-output change relative to the corresponding unchanged parent, in meV/atom.

Parent	Hydrogen readout	Graph (%)	Joint (%)	Energy change
FM	None	25.31	24.17	0
FM	Radial rule	27.29	25.94	+1.89
FM	Conditional flow	27.40	26.04	−13.32
GAGA	None	23.85	23.54	0
GAGA	Radial rule	24.06	23.75	+1.05
GAGA	Conditional flow	24.06	23.75	−2.04

For FM, the learned readout lowers energy by 15.21 meV/atom relative to the radial rule (95% interval for the reduction [10.10, 21.71]), with reductions in both fits. Graph and joint yield each differ by only +0.10 percentage points ([−0.21, 0.42]). Relative to the unchanged parent, graph validity increases by 2.08 points ([1.15, 3.13]) and joint yield by 1.88 points ([1.04, 2.71]). Every previously graph-valid parent remains unchanged. Allowing all hydrogens to move, rather than only the detached ones, further reduces energy by 10.25 meV/atom ([6.96, 14.72]), with identical graph and joint counts. This ablation supports learning their arrangement together. GAGA also benefits physically: its learned readout lowers energy by 3.10 meV/atom relative to the radial rule ([1.04, 6.34]).

With the same learned readout for both generators, FM exceeds GAGA in graph validity by 3.33 points ([0.52, 6.15]). The joint difference is +2.29 points ([−0.63, 5.21]), so this physical-quality ordering remains uncertain. The all-output energy averages include invalid geometries. As a descriptive check on valid outputs, 261 paired FM structures pass graph validation under both readouts; 18 are newly repaired. On these 18 pairs the learned readout has 83.81 meV/atom lower mean energy than the radial rule. This conditional subset is not used for model selection or for the primary energy interval.

The confirmation generates 1,920 parent trajectories and 5,760 derived outputs. It requires 2,460 new GFN2 calculations: one per parent and 540 unique changed geometries. Identical coordinates reuse the original calculation. These costs are separate from development, which uses 2,560 derived outputs and 155 new GFN2 calculations, and from fitting the hydrogen models. All sampled outputs remain in the reported denominators. The result establishes a local physical benefit of the readout, not sampling from a global Boltzmann distribution.

O COMPLETE GENERATORS ON COMMON COMPOSITIONS

We evaluate the complete systems on the same 16 new compositions, eight each with 17–28 and 29–40 atoms. The panel is fixed by metadata hashes, excludes prior confirmation panels, and is absent from both processed generator corpora. All conditions are neutral organic singlets within the GAGA adaptation’s element vocabulary. Each system generates 16 raw structures per composition and run, for 512 scored outputs per row of Table 23.

This comparison answers a different question from the shared-EGNN experiment. The FlowMol variants use a 5.90-million-parameter inherited backbone; the GAGA models use the 2.38-million-parameter EGNN trained from scratch on 20,000 OMol25 structures. Their architectures, training histories, physical supervision, and costs differ. GAGA uses its validation-selected truncation at 650, evaluated at both 128 calls and the complete 651-call schedule. The paired-update model keeps its

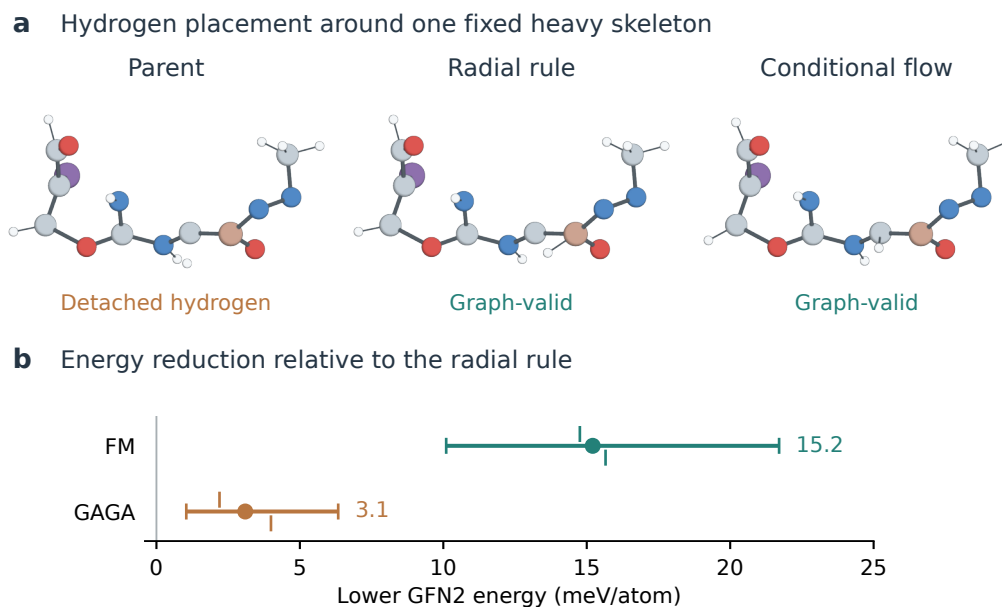


Figure 9: A conditional hydrogen flow learns more than attachment distance. Top: the same saved FM output before readout, after the radial rule, and after the learned flow. All atoms are shown with a shared camera and scale. Sticks are distance contacts in every panel, not inferred chemical bond orders. The example is selected by a fixed metadata hash among repaired, jointly graph-valid examples with at most 28 atoms; energy is not a selection criterion. Bottom: all-output energy reductions relative to the radial rule, with 95% composition-bootstrap intervals and the two fit-specific means. Carbon, hydrogen, nitrogen, oxygen, boron, and iodine are gray, white, blue, red, tan, and purple.

two original continuations. The correction-head study uses two head-fitting seeds on one shared parent. Intervals condition on these fitted systems and resample the 16 compositions.

Same compositions · raw generated coordinates

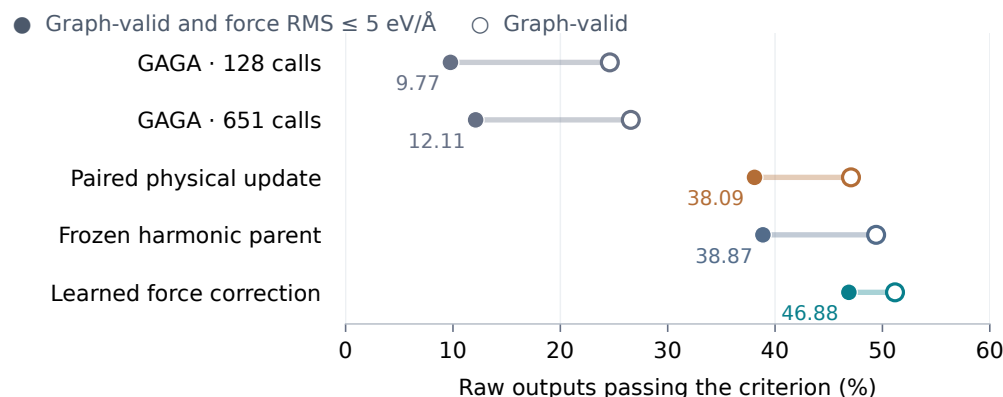


Figure 10: Graph support and physical quality on the same compositions. Open circles count graph-valid outputs; filled circles also require GFN2 force RMS at most 5 eV/Å. The gap shows graph-valid outputs above that force threshold. Every denominator includes all 512 attempts. The models have the distinct training histories and inference costs described in the text.

The contextual head was the prespecified primary challenger. It improves joint yield over its parent by 7.81 percentage points, with a 95% interval of [4.88, 10.94]; the run-specific gains are 7.03 and 8.59

Table 23: Raw outputs in the common-panel study. Force medians use each model’s own graph-valid outputs with successful GFN2 calculations. Joint yield uses all attempts. “Calls” separates backbone and correction-head evaluations.

Model	Calls	Graph (%)	Force (eV/Å)	Joint (%)
GAGA	128	24.61	5.408	9.77
GAGA, complete schedule	651	26.56	5.176	12.11
Paired physical update	128	47.07	3.581	38.09
Paired update, no terminal noise	128	46.88	2.929	40.04
Frozen harmonic parent	128+64	49.41	3.394	38.87
Small correction, 2,000 updates	128+64	50.00	3.077	43.16
Small correction, 20,000 updates	128+64	51.17	2.854	46.88
Wide correction, 20,000 updates	128+64	50.20	2.831	46.09
Contextual correction, 20,000 updates	128+64	50.78	2.864	46.68

points. Its differences from the small and wide 20,000-update heads are -0.20 points ($[-1.95, 1.56]$) and $+0.59$ points ($[-1.17, 2.34]$). Thus the better internal force-target prediction of the contextual architecture does not produce an established generation advantage over either simpler control.

Relative to GAGA at 128 and 651 calls, the contextual complete system gains 36.91 points ($[26.56, 46.09]$) and 34.57 points ($[24.80, 43.95]$). The original paired-update system also exceeds both GAGA settings, by 28.32 points ($[20.70, 35.74]$) and 25.98 points ($[18.75, 33.40]$). These are complete-system results on common conditions. The shared-backbone results in Appendices K.1 and N control architecture and training exposure. Learned correction improves both EGNN generators under equal physical supervision, with uncertain ordering between them.

Training duration and terminal noise. The published-model and terminal-noise controls were designated before reading the new GFN2 comparison. The zero-noise output is captured from the same neural trajectory as the published sampler, before its final $0.025 \text{ \AA}$ perturbation. Removing that noise increases joint yield by 1.95 points ($[0.39, 3.71]$). The small learned correction exceeds the paired-update model by 8.79 points ($[3.91, 13.67]$), and its zero-noise version by 6.84 points ($[1.95, 11.72]$). Their physical-training sets and parent histories differ, so this comparison combines those choices with the adaptation mechanism.

After inspecting the main confirmation, we add a training-duration ablation using the already-saved 2,000-update prefix of each small-head fit. All targets, initial weights, training-state choices, and generation settings match the 20,000-update model. Longer fitting adds 3.71 points ($[1.76, 5.86]$), with positive changes in both runs. This supplemental ablation is a later diagnostic, not an independently chosen confirmation panel.

Across all unoptimized outputs, the small 20,000-update head also lowers GFN2 energy by a mean 0.0951 eV/atom relative to its matched parent. The signed-change 95% interval is $[-0.2022, -0.0364] \text{ eV/atom}$, and both runs decrease. This readout includes invalid outputs and measures the change across paired generator attempts, rather than equilibrium energy differences between isomers.

Additional graph diagnostics. Requiring RDKit to assign zero radical electrons gives joint yields of 46.29% for the small correction, 46.09% for the contextual correction, and 9.57% and 11.52% for the two GAGA budgets. The relative result therefore also appears under this stricter graph-assignment check. Within each composition and run, all graph-valid outputs have distinct inferred labeled connectivity graphs. This 16-draw diagnostic measures observed duplication, not coverage of the possible isomers or their equilibrium probabilities.

The six-system confirmation contributes 3,072 neural trajectories and GFN2 readouts. Its parameter-matched control adds 40,000 head updates; the preceding small/contextual capacity study used 80,000 updates. The supplemental published and short-fit study adds 1,024 neural trajectories and 512 further outputs obtained by sharing the published model’s trajectory across the two noise settings, with 1,536 GFN2 readouts. The physical teacher is reused throughout. No generated coordinates are optimized.

O.1 LEARNING FROM PHYSICALLY CORRECTED REFERENCES

We test a second way to supply physical training targets: correcting OMol25 reference geometries before fine-tuning. This separates the value of physical labels from the effect of additional training on a small reference set.

Matched reference sets. From the existing 3,000-row training block, we select 128 distinct compositions by fixed metadata hashes, with 32 in each of the 8–16, 17–24, 25–32, and 33–40 atom ranges. Each contributes one reference structure. We construct a corrected copy using a bounded Cartesian L-BFGS search under the inversion-averaged eSEN potential. All coordinates may move, while geometric validity and the inferred labeled graph are preserved. Each search allows at most 96 steps and 160 energy/force evaluations, with a 0.05 Å per-atom step cap. It stops early when the largest atomic force is at most 0.1 eV/Å.

Every composition remains in both sets. If a search stalls or reaches its budget, we retain its last accepted point and record its termination status. Of the 128 searches, 110 meet the force stopping criterion and 124 lower force RMS. Including inversion pairs, target preparation uses 6,576 eSEN evaluations. Independent GFN2 single points evaluate all 256 original and corrected structures.

Table 24: Quality of the 128 training references before and after physical correction. These are training labels, not generated outputs. GFN2 scores are used only for evaluation.

Metric	Original references	Corrected references
Median eSEN force RMS (eV/Å)	2.005	0.050
Median GFN2 force RMS (eV/Å)	1.756	0.397
GFN2 force RMS ≤ 1 eV/Å (%)	22.66	89.84

Direct flow-matching adaptation. Two fine-tuning runs start from the same 5.90-million-parameter harmonic-source checkpoint. Within each run, one student learns from the original references and another from their corrected copies. Each receives 2,000 updates at batch size one. They share reference indices, source draws, rotation and atom-matching randomness, optimizer settings, and 0.005 Å centered endpoint smoothing. Composition frequencies are uniform and identical in the two target sets. This experiment uses direct FM training without parameter subtraction; its target is the specified empirical reference distribution.

The confirmation panel contains 16 previously ungenerated compositions, eight each with 17–28 and 29–40 atoms, absent from both processed corpora and earlier generation panels. Each model generates 16 samples per composition and run. All three models use 128 backbone calls and zero terminal noise. The original parent is regenerated with the same settings. No checkpoint is selected using these outcomes, and no generated structure is physically optimized.

Table 25: Raw generation on the reference-target confirmation panel, with 512 attempts per model. GFN2 joint yield requires chemical-graph validity and force RMS at most 5 eV/Å, divided by all attempts. Force medians use each model’s own graph-valid outputs with successful calculations.

Model	Graph validity (%)	Median force (eV/Å)	Joint yield (%)
Unadapted parent	38.09	3.083	31.64
Original-reference fine-tuning	30.47	2.762	27.34
Corrected-reference fine-tuning	39.45	2.558	33.01

Corrected references improve joint yield over matched original-reference fine-tuning by 5.66 percentage points, with a composition-bootstrap 95% interval of [2.54, 8.59]. The run-specific gains are +8.20 and +3.13 points. Relative to the unadapted parent, the pooled gain is smaller: +1.37 points ([−3.52, 6.05]), with run-specific changes of −1.95 and +4.69 points. Thus physical labels help compared with the same additional reference training, while an overall advantage over the parent remains uncertain. The intervals condition on these two fitted runs. This study adds 8,000 optimizer updates, 1,536 raw neural outputs, and 1,536 GFN2 generation readouts, in addition to target preparation and its 256 GFN2 label checks.

Physical evaluation of the frozen source models. This evaluation concerns the pretrained source models on the 10-composition additional test set. We score every generated output and its original reference with the same conservative eSEN checkpoint at the original charge and spin. Energies and forces are averaged over inversions as $E_+(X) = [E(X) + E(-X)]/2$ and $F_+(X) = [F(X) - F(-X)]/2$. We pair differences within composition before averaging. Per-atom energies also subtract the composition’s reference energy; that reference cancels in a paired model difference. The evaluation uses 5,160 raw energy/force queries, including both inversions of references in each fine-tuning run.

For each sample, force RMS is the square root of the mean squared atomic force norm; we then average this value over samples. The jointly valid comparison retains only pairs accepted under both models, with 3–35 pairs in each composition/run combination. The all-attempt comparison also includes graph-invalid outputs. Reporting both distinguishes overall scores from scores on pairs that remain valid under both methods.

Most of the aggregate improvement comes from run 1. The all-attempt energy differences are -0.08049 and -0.00070 eV/atom in runs 1 and 2, respectively; the force-RMS differences are -1.21996 and $+0.29302$ eV/Å.

The original endpoint checkpoint. We also evaluate the original 30,000-step OMol25 checkpoint on coordinate generation, fixing the composition and neutral/no-bond inputs. Its endpoint sampler uses a first-step bootstrap and 127 Euler updates, for 128 network calls. Keeping the model’s predicted categorical history gives 106/640 and 113/640 graph-valid outputs in two sampling streams. Fixing the history categories and centering the initial bootstrap gives 116/640 and 120/640. Both streams use the same checkpoint; the second variant changes both history handling and bootstrap centering. LatentSpring’s source models additionally receive shared task adaptation and 3,000 specialized updates. The matched Gaussian and covariance-Gaussian fine-tuning runs therefore provide the controlled tests of source choice.

P NORMALIZATION OF THE HARMONIC MIXTURE

We first compute the density for one tree, then average over trees. The key is the change of coordinates from independent edge vectors to centered atom positions. For a tree T , let $B_T \in \mathbb{R}^{(N-1) \times N}$ be its edge-incidence matrix. Each row has $+1$ and -1 at the endpoints of one edge, so $B_T X$ contains the edge displacements. Let $U \in \mathbb{R}^{N \times (N-1)}$ have orthonormal columns that span the centered subspace for one spatial dimension.

Because $B_T \mathbf{1} = 0$, we have $(B_T U)(B_T U)^\top = B_T B_T^\top$. By Cauchy–Binet, $\det(B_T B_T^\top)$ is the sum of the squares of its N maximal minors. Every maximal minor of a tree incidence matrix is ± 1 , so this determinant is N . The absolute Jacobian is therefore $\sqrt{N}$ in one spatial dimension and $N^{3/2}$ in three dimensions. The density for a fixed tree is consequently $q(X \mid T, c) = N^{3/2} \prod_{e \in T} h_e(B_T X)$, with each edge density evaluated at its corresponding row of $B_T X$. It integrates to one. Averaging these normalized components with Equation (1) gives Equation (3) and preserves normalization.

Q SOURCE INVARIANCE UNDER SYMMETRY COUPLING

Alignment chooses a rotation and atom relabeling using the reference structure. We show why a subsequent shared random augmentation restores the source marginal, even when the source is not Gaussian.

Let G be the compact group of proper rotations and permutations of identical conditioning labels. Assume that q_0 is invariant under G and $X \sim q_0$. The pairing procedure selects $g(X, Y) \in G$ using the target Y . Independently draw H from the uniform, or Haar, distribution on G . Conditional on the orbit of X and the selected g , the transformed source HgX has the same uniform orbit distribution as HX , by Haar invariance. An orbit here is the set of configurations obtained by applying these symmetries to one configuration. Alignment changes neither the orbit nor its probability. Averaging over source orbits therefore recovers q_0 .

Table 26: Harmonic-minus-Gaussian eSEN scores on the frozen models’ 10-composition test. Negative values favor harmonic. Rows distinguish all attempts from paired outputs that are both graph-valid. Brackets give paired 95% confidence intervals.

Metric	Outputs	Difference	Interval
Energy (eV/atom)	All attempts	−0.0406	[−0.0579, −0.0232]
	Both outputs valid	−0.0340	[−0.0692, −0.0003]
Force RMS (eV/Å)	All attempts	−0.4635	[−0.6782, −0.2512]
	Both outputs valid	−0.5389	[−0.8950, −0.2089]

The paired endpoints can still be correlated. The training objective uses their conditional displacement target directly; this argument establishes the source marginal, not independence of the endpoints.

R COVARIANCE CONTROLS

The shell-tree control uses the same radius-based edge scale but concentrates edge lengths around it. For each edge, it samples a length $R = \ell \exp(sZ - s^2/2)$, where $Z \sim N(0, 1)$ and $s = 0.2$, then samples a uniform direction. Since $\mathbb{E}R^2 = \ell^2 \exp(s^2)$ and the direction has second moment $I/3$, the edge covariance is $vI = \ell^2 \exp(s^2)I/3$.

Let A_T map edge coordinates to centered atom coordinates. Independence of edges gives $C_T = A_T \text{diag}(v_e) A_T^\top$, where C_T is the atom-position covariance for one spatial axis and the diagonal matrix contains edge variances. Replacing the shell vectors by independent Gaussians with covariance $v_e I$ leaves C_T unchanged. All tree components have zero mean, so the mixture covariance $\mathbb{E}_T C_T$ is also unchanged.

The single-Gaussian control estimates this mixture covariance by averaging a finite number of tree covariances, with randomness seeded by composition. We then average over permutations within each atom species. This preserves positive definiteness on the centered subspace and makes the covariance invariant to relabeling identical atoms. The resulting Gaussian is exactly normalized for its chosen covariance, but the covariance only approximates that of the full tree mixture.

S JARZYNSKI IDENTITY FOR ESCORTED TRANSPORT

This appendix states the change-of-variables identity behind our work-corrected local targets. Jarzynski’s relation connects exponential work averages to free-energy differences, including in single-molecule pulling (Jarzynski, 1997; Hummer & Szabo, 2001). Escorted transport adds an imposed coordinate flow and accounts for its volume change (Vaikuntanathan & Jarzynski, 2008). Here, “work” denotes the corrected energy-like quantity whose exponential supplies an importance weight.

Fix a composition and use volume measure on its centered coordinate space. Let x collect the independent coordinates and let $\beta = 1/(k_B T_{\text{phys}})$. The unnormalized target is $\rho_1(x) = \mathbf{1}_\Omega(x) \exp[-\beta E(x)]$, with finite, nonzero $Z_1 = \int_{\mathcal{H}_N} \rho_1(x) dx$. The allowed region Ω can, for example, confine the molecule so that this integral is finite. The indicator is one inside this region and zero outside it; the normalized target is ρ_1/Z_1 .

Let $q_0 > 0$ be the normalized source in Equation (3) and let $\Phi : \mathcal{H}_N \rightarrow \mathcal{H}_N$ be a differentiable, invertible map with nonzero intrinsic Jacobian. Its absolute determinant $|\det D\Phi|$ measures local expansion or compression in the independent coordinates. For $\Phi(x_0) \in \Omega$, define

$$W_\Phi(x_0) = E(\Phi(x_0)) + \beta^{-1} \log q_0(x_0) - \beta^{-1} \log |\det D\Phi(x_0)|, \quad (13)$$

and set $W_\Phi = +\infty$ outside the allowed region, giving zero weight there. The source-density and Jacobian terms correct for where the mapped samples came from. Multiplying the exponential weight by the source density and changing variables gives

$$\mathbb{E}_{q_0}[e^{-\beta W_\Phi}] = \int_{\mathcal{H}_N} \rho_1(\Phi(x_0)) |\det D\Phi(x_0)| dx_0 \quad (14)$$

$$= Z_1. \quad (15)$$

Equivalently, the auxiliary source energy $E_0 = -\beta^{-1} \log q_0$ has $Z_0 = 1$ in this measure, yielding a Jarzynski-type normalizer identity. For an observable f , we estimate the target mean with $\sum_j w_j f(\Phi(x_j)) / \sum_j w_j$, where $w_j = e^{-\beta W_\Phi(x_j)}$. This is self-normalized importance sampling. Its finite-sample variance depends on how well the mapped proposals cover the target.

For an exact, sufficiently regular ordinary differential equation (ODE) flow, the intrinsic log determinant equals the integrated velocity divergence. Our implemented work corrections instead use explicit maps with tractable local densities: Gaussian translations in Appendix T and periodic rotor maps with analytic Jacobians in Appendix U. We do not use an exact neural-flow likelihood to evaluate the reported generator. Its outputs use the midpoint solver and terminal noise in Section 4.

T LOCAL FORCE-ESCORTED WORK DISTILLATION

The physical teacher constructs nearby candidate structures and determines how often each is used as a flow-matching target. We compare uniform sampling of force-shifted candidates with energy-weighted training (Dern et al., 2025; Midgley et al., 2023). A normalized Gaussian around each anchor makes the complete escorted-work weights explicit (Vaikuntanathan & Jarzynski, 2008).

Energy, force, and mirror symmetry. The energy model gives a scalar $E(X)$ and forces $F(X) = -\nabla_X E(X)$. Each atom’s force points toward decreasing energy. To treat mirror-related structures consistently, we evaluate both X and its inversion $-X$ and use $E_+(X) = [E(X) + E(-X)]/2$ and $F_+(X) = [F(X) - F(-X)]/2$. The minus sign in the force average ensures $F_+ = -\nabla E_+$. Inverted structures thus receive equal energies and opposite forces. Before constructing a displacement, we subtract the mean atom force so the displacement stays in the centered coordinate space.

Local reference and target. For a graph-valid generated anchor A , let G_A be its inferred labeled graph and define $q_A(X) = \mathcal{N}_{\mathcal{H}_N}(A, \sigma_A^2 I)$. We use thermal energy $kT = k_B T_{\text{phys}} = 0.0258519998$ eV at 300 K and set

$$\sigma_A = \min \left(0.03 \text{ \AA}, \sqrt{\frac{0.10 \text{ \AA} kT}{\|F_+(A)\|}} \right), \quad \delta_A = \frac{\sigma_A^2}{kT} F_+(A). \quad (16)$$

The norm is the Frobenius norm across all atom-force components. These choices bound the displacement norm by 0.10 Å and the Gaussian width by 0.03 Å. At zero force, we use the maximum width. Both quantities are fixed before sampling $X \sim q_A$. We then translate each candidate to $Y = X + \delta_A$. This map stays within the centered subspace and has determinant one.

The unnormalized local target is

$$\rho_A(Y) = q_A(Y) \exp \left[-\frac{E_+(Y) - E_+(A)}{kT} \right] \mathbf{1}_{G_A}(Y). \quad (17)$$

The Gaussian keeps structures near the anchor, while the energy factor favors lower-energy structures. The indicator restricts candidates to the anchor’s inferred labeled bond orders, formal charges, and radical counts, together with the connected/nonoverlap checks. Thus this is a restrained local distribution, not an unrestricted equilibrium distribution over molecules.

Why the force displacement has this form. Approximate energy near the anchor by $E_+(Y) - E_+(A) \simeq -\langle F_+(A), Y - A \rangle_F$. The inner product sums force times displacement over all atoms and coordinates. Multiplying this linearized Boltzmann factor by the Gaussian and completing the square gives

$$q_A(Y) \exp \left[\frac{\langle F_+(A), Y - A \rangle_F}{kT} \right] \propto \mathcal{N}_{\mathcal{H}_N} \left(Y; A + \frac{\sigma_A^2}{kT} F_+(A), \sigma_A^2 I \right). \quad (18)$$

Before applying the graph restriction, the force translation therefore samples the energy-weighted Gaussian exactly under the linear approximation. Equivalently, the Gaussian restraint is a virtual harmonic spring with stiffness kT/σ_A^2 , and the force shifts its mean by force divided by stiffness. The same-graph check then truncates this distribution.

Complete local work. Uniform weighting uses the linear approximation above. Complete-work weighting instead evaluates the nonlinear energy to correct departures from that approximation. With reference Hamiltonian $H_0 = -kT \log q_A$ and target Hamiltonian $H_1 = H_0 + E_+ - E_+(A)$ plus the hard support restriction, the work is

$$W_A(X) = E_+(Y) - E_+(A) + kT[\log q_A(X) - \log q_A(Y)] \quad (19)$$

on allowed candidates, and $+\infty$ otherwise. The log-density terms correct for the shift from the proposal X to Y . Change of variables gives $\mathbb{E}_{q_A}[\exp(-W_A/kT)] = \int \rho_A(Y) dY$. Analytic checks recover constant work for a constant-force translation, and the normalizer and weighted mean for a quadratic energy. Substituting the linearized energy in Equation (18) gives $W_A = -\sigma_A^2 \|F_+(A)\|_F^2 / (2kT)$ for every allowed candidate. Their weights are therefore equal in that approximation, explaining the less expensive force-only teacher.

Training pools and controls. Each fine-tuning run selects eight distinct training compositions with 17–28 atoms using a fixed metadata hash, then generates 32 anchors per composition. These compositions are excluded from evaluation. Invalid anchors count as generation attempts but do not enter the teacher pool. Each valid anchor produces eight force-shifted candidates; candidates outside its support receive zero weight. The complete-work model samples training endpoints with self-normalized $\exp(-W_A/kT)$ weights. The unweighted-escort model samples uniformly from the same allowed candidates. The replay model uses the original retained anchors, while the frozen model receives no further training. Anchors with no allowed candidate are recorded and excluded from all student pools in the same way.

Each student alternates between 500 FM updates on original references and 500 updates on generated endpoints. Initialization, anchor draws, optimization, and sampling budgets are shared. Replay needs no force queries; the unweighted escort needs anchor forces; complete work additionally needs energies at valid candidate endpoints. We distinguish the queries used in the shared research experiment from those required by each method on its own. Generation uses only the final network. Its temperature feature is fixed during training and evaluation; 300 K sets the local teacher weights.

Self-normalized weights introduce finite-particle bias. We report effective sample size (ESS) over each anchor’s eight candidates: $(\sum_j w_j)^2 / \sum_j w_j^2$. Equal weights give ESS equal to the candidate count, while a single dominant candidate gives ESS close to one.

Direct fine-tuning results. We train three students for 1,000 updates in each run. Across 512 generated anchors, 351 pass the graph checks and 349 have at least one allowed force-shifted candidate. Composition-level mean ESS is about 1.27–1.87 out of 8. The weights are therefore concentrated on few candidates, limiting resolution of the local target with this pool size.

Among paired outputs that are both graph-valid, complete-work training gives lower energy than unweighted-escort training by 0.00758 eV/atom (paired 95% interval $[-0.01424, -0.00113]$ for the signed difference). Force RMS is lower by 0.16174 eV/Å ($[-0.27660, -0.05590]$). Both runs favor complete work, and the all-output comparison has the same direction.

However, all three directly fine-tuned students perform worse than the frozen model. Complete work has 55.16% graph validity versus 62.11% for the frozen model, and energy on paired, jointly valid outputs is *higher* by 0.10587 eV/atom ($[0.09009, 0.12168]$). Replay also worsens performance after training on the mixed targets. These results motivate subtracting the matched replay update rather than using the physical student directly.

Teacher preparation uses 6,232 raw energy/force queries: 702 at anchors and 5,530 at allowed candidates. The unweighted escort could omit candidate-energy queries, and replay could omit all physical queries. Scoring the outputs of the four models uses a further 10,240 queries, separate from generation cost.

Applying the paired physical update. Instead of deploying the directly fine-tuned student, we use parameter arithmetic (Ilharco et al., 2023) to subtract its matched replay update:

$$\theta_{\text{final}} = \theta_{\text{base}} + (\theta_{\text{physical}} - \theta_{\text{replay}}), \quad (20)$$

for both unweighted-escort and complete-work students. The coefficient is fixed at one. We change parameter tensors only; buffers and the source distribution remain fixed.

Table 27: Graph-valid outputs after direct physical fine-tuning, out of 640 attempts per run. The frozen model receives no updates; all other variants have the same additional training budget.

Run	Frozen	Replay	Unweighted escort	Complete work
1	418	340	330	344
2	377	345	371	362

Table 28: Paired updates evaluated after the direct fine-tuning experiment. Energy changes use paired, jointly valid outputs and are relative to the frozen model, with paired 95% confidence intervals. Counts show the two generation streams.

Model	Graph-valid outputs (two streams)	Energy change (eV/atom)
Frozen	418/640; 377/640	0
Paired force update	407/640; 387/640	$-0.01554 [-0.02084, -0.01031]$
Paired work update	406/640; 382/640	$-0.01604 [-0.02188, -0.01013]$

Both paired variants lower energy and force RMS in both runs. For complete work, force RMS on paired, jointly valid outputs falls by $0.38403 \text{ eV/\AA}$ (interval $[-0.48124, -0.28932]$); all-attempt energy and force scores also improve. Its graph-validity difference from the frozen model is -0.55 percentage points, with interval $[-2.58, 1.56]$.

The complete-work and unweighted paired variants have similar scores. Complete work minus unweighted gives -0.00243 eV/atom on jointly valid pairs ($[-0.00756, 0.00264]$) and $+0.00022 \text{ eV/atom}$ over all attempts ($[-0.00557, 0.00649]$). Force and graph-validity intervals also include zero. Teacher preparation requires 702 anchor queries for the unweighted variant versus 6,232 for complete work. Both retain the 128-call neural sampler.

Evaluating these two paired variants adds 2,560 generated outputs and 5,120 physical queries. Combining parameters requires no optimizer steps. Based on these development results, we select the force-only update for the new-composition evaluation in Section L.2.

U MOLECULAR ROTOR WORK PROTOCOL

We use a controlled one-dimensional motion to test the work correction on molecular energy curves. The three training molecules are $\text{C}=\text{C}(\text{N})\text{N}(\text{C})\text{C}(\text{N})=\text{O}$, $\text{CCOC}(\text{C})(\text{C})\text{N}$, and $\text{CC}(=\text{O})\text{NCC}=\text{O}$. We inspect the first five rows of an existing training subset and take the first three distinct compositions with a methyl rotor that preserves the inferred labeled graph on a 64-angle grid. Selection uses no energy queries. Only the methyl hydrogens rotate around the carbon-anchor axis; other internal coordinates remain fixed. We remove overall translation and define densities with respect to the periodic angle measure $d\phi$ on $[-\pi, \pi)$.

Reference, target, and coordinate map. The starting density is $q_0(\phi) = \exp(\cos \phi)/(2\pi I_0(1))$, with normalizing integral $2\pi I_0(1) = \int_{-\pi}^{\pi} \exp(\cos u) du$. This is a circular reference distribution, distinct from the coordinate source used by the neural generator. Its reference Hamiltonian is $H_0 = -kT \log q_0$, and the target Hamiltonian is $H_1 = H_0 + U - U(0)$. We transform angles with $\psi_a(\phi) = \phi + a \sin \phi$ for $a \in \{-0.5, 0, 0.5\}$. The derivative, $J_a(\phi) = 1 + a \cos \phi$, is the local volume-change factor. Since $|a| < 1$, this map is smooth and invertible on the circle. Its complete work is

$$W_a(\phi) = U(\psi_a(\phi)) - U(0) + kT [\log q_0(\phi) - \log q_0(\psi_a(\phi)) - \log J_a(\phi)]. \quad (21)$$

The density terms correct the changed angular sampling distribution, and the Jacobian term corrects expansion or compression of angular intervals. Changing variables gives

$$\mathbb{E}_{q_0} e^{-W_a/kT} = \int_{-\pi}^{\pi} q_0(y) e^{-[U(y)-U(0)]/kT} dy = Z_1/Z_0, \quad Z_0 = 1. \quad (22)$$

Numerical reference. All 1,536 single-point calculations succeed. For each rotor, we construct a periodic cubic interpolation of 512 GFN2 energy values. The 256-knot interpolation has error at most 10^{-8} eV at the additional midpoint samples in this study. Increasing the reference quadrature from 32,768 to 65,536 points changes free energy by less than 10^{-10} eV. These checks assess interpolation and numerical integration errors for the GFN2-based reference curve.

Estimators and finite-sample effects. We compute total variation on 64 equal angular bins, using dense quadrature for the target probabilities. Self-normalized estimates of observables have finite-sample bias, whereas the average unnormalized weight is an unbiased estimator of the normalizing constant. The free-energy estimate is $-kT \log \hat{Z}$; its RMSE includes the bias introduced at finite sample size. At 2,048 draws, complete-work ESS ranges from 788 to 1,892 across the six nonidentity cases. Some incomplete weights have higher ESS despite targeting the wrong distribution. ESS alone therefore does not establish correct target recovery.

U.1 COMPLETE-WORK CORRECTION ON MOLECULAR ROTORS

For this test, one angle ϕ specifies a family of three-dimensional structures: the three hydrogens of a methyl group rotate around its bond axis while the other internal coordinates remain fixed. We select three such rotors from training molecules in fixed data order using geometry alone. A periodic interpolation of 512 GFN2 energies defines $U(\phi)$ for each rotor, with $U(0)$ as the reference energy. The target is $\rho(\phi) \propto q_0(\phi) \exp[-(U(\phi) - U(0))/kT]$, where q_0 is a normalized von Mises distribution with concentration one. It favors angles near zero; the energy factor reweights those angles while retaining this restraint. Here $kT = k_B T_{\text{phys}}$ at 300 K.

We apply three fixed angle maps, including the identity map, to the starting samples. For each map, we compare complete-work weighting, energy-only weighting, weighting without the Jacobian correction, and no weighting. Complete work accounts for both the reference-density change and the expansion or compression of angle intervals. Each setting uses 128 repetitions at sample counts from 8 to 2,048. The maps and work formula are specified above in Appendix U.

Total variation measures the discrepancy between estimated and target angular probabilities; zero indicates agreement. Free-energy RMSE measures error in the logarithm of the target normalizer, expressed in energy units.

At 2,048 samples, complete work has lower mean total-variation error than both incomplete-weight controls in all six nonidentity cases. The average error is 0.0636 for complete work, 0.1532 for energy-only weights, and 0.1134 without the Jacobian. Complete work therefore reduces this error by 58.5% relative to energy-only weighting. Its free-energy RMSE ranges from 0.158 to 0.734 meV. Independent density-ratio quadrature also recovers the target normalizer in all nine molecule/map cases. With the identity map, the three weighted methods coincide. These results test the full work correction on restrained molecular energy curves, rather than equilibrium sampling by the neural generator.

V PHYSICAL QUALITY OF THE MATCHED GENERATORS

We evaluate physical scores for the 16,384 saved outputs from the independent EGNN comparison in Section I.3. These are the original 128-call outputs; no models are retrained and no coordinates are changed. We fix the physical-evaluation protocol before computing these scores. eSEN evaluates every output and its reflection. GFN2 evaluates every graph-valid output and both inversions of each reference. All 2,198 GFN2 calculations succeed, including 256 reference checks that pass inversion tests. eSEN uses 33,024 evaluations.

Joint quality yield divides outputs that pass both graph validity and a physical threshold by *all generation attempts*. Graph-invalid structures therefore remain in the denominator. Figure 12 sweeps the force-RMS threshold instead of treating one cutoff as a universal definition of physical validity. Passing a threshold does not certify a stationary geometry. These models share the architecture and training budget in Appendix K; none receives a physical update in this source/objective comparison.

a Controlled methyl rotation on three molecules

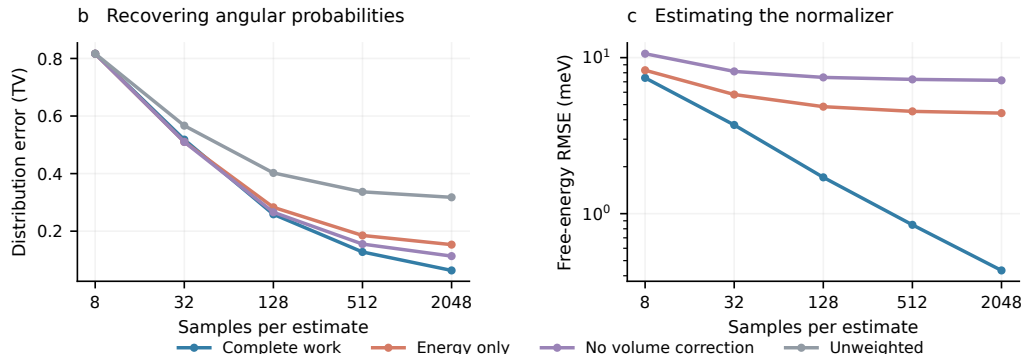
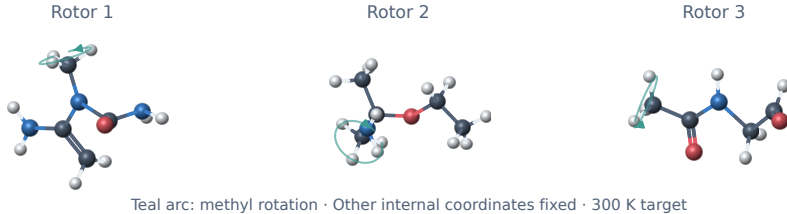


Figure 11: Work correction on controlled molecular rotors at 300 K. (a) Reference structures and the prescribed methyl rotations. (b,c) Angular-distribution and normalizer errors averaged over three molecules and two nonidentity maps, with 128 repetitions per case. Complete weights account for the proposal bias left by energy-only and incomplete weights. Unweighted samples have no free-energy estimator. Each target retains its angular reference restraint.

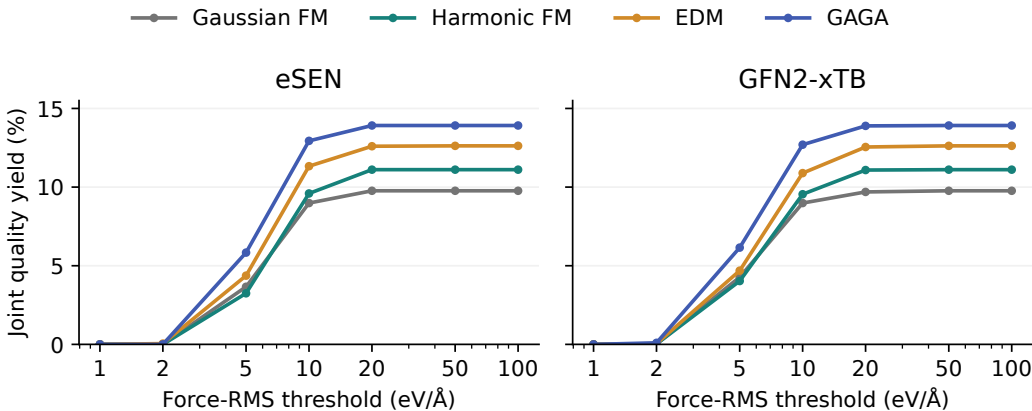


Figure 12: Fraction of all attempts that are graph-valid and have force RMS below the indicated threshold. Curves pool both independent initializations and all 64 compositions. GAGA has the highest pooled yield at most nontrivial thresholds under both potentials.

Higher graph validity from the harmonic source does not, by itself, imply better physical scores among valid outputs. GAGA has stronger pooled physical results in this matched setting. The separate physical-adaptation experiments test the effect of local physical targets on trained generators.

Table 29: GFN2 scores for the matched generators. Energy and force are medians across each method’s own graph-valid outputs; yield uses all 4,096 attempts per method. Energy excess is relative to one OMol25 geometry of the same composition. The generated bond graph may differ from that reference, so this is not a same-graph strain-energy measurement.

Method	Energy excess (eV/atom)	Force RMS (eV/Å)	Yield at 5 eV/Å (%)
Gaussian FM	0.6820	5.2911	4.27
Harmonic FM	0.7114	5.8183	4.03
EDM	0.6799	5.8768	4.69
GAGA	0.6604	5.2785	6.15

W TRAINING-COMPONENT CONTROLS

W.1 WHAT SELF-CONDITIONING CONTRIBUTES

We remove self-conditioning from the harmonic-source model while keeping its initialization, 3,000 training structures, source distribution, and pairing procedure unchanged. One single-pass model receives 3,000 optimizer updates. A second receives 6,000, matching the self-conditioned model’s number of backbone training evaluations. Both single-pass models use 64 midpoint steps at generation, so every model uses 128 network calls per output.

Self-conditioning improves graph validity by 7.23 percentage points at equal optimizer updates (95% interval [4.69,9.77]) and by 12.43 points at equal training forward passes ([9.70,15.23]). Both runs improve in both comparisons. Matching pass counts does not match every operation or wall-clock cost. This control asks whether additional single-pass training recovers the validity gain; it does not in these runs. The update-direction control is reported in Section L.1.

Contribution of the force displacement. We also remove only the force shift from local training targets. Gaussian perturbations, candidate indices, widths, training schedule, and the paired parameter update remain unchanged. Relative to this control, force-shifted targets lower energy on paired, jointly valid outputs by 0.01939 eV/atom (95% interval [-0.02417,-0.01455] for the signed difference) and force RMS by 0.3926 eV/Å ([-0.4696,-0.3263]). Both runs improve. Graph validity remains similar: force minus control is +0.39 percentage points, with interval [-1.33,2.19]. Because the original force-dependent widths and candidate-selection rule remain fixed, this ablation tests the displacement itself, not the removal of all force information.

W.2 MATCHED EXPERIMENT ON SOURCES AND PHYSICAL ADAPTATION

To complete the four-setting comparison, we construct physical targets from the Gaussian model’s own outputs using the same procedure as for harmonic. In each run, the frozen Gaussian model generates 32 anchors for each of the same eight training compositions. We record all 256 attempts and discard invalid anchors without replacing them. We retain the eight-candidate force-escort procedure, temperature, perturbation bounds, source seeds, and 1,000-update training schedule. For each physical student, we train a matched replay student from the same Gaussian base and add their parameter difference to that base with coefficient one. Within each physical/replay pair, reference examples, source randomness, training times, and teacher-selection draws are paired.

The two Gaussian runs retain 160 and 138 anchors after candidate-support checks. Teacher preparation uses 5,276 raw eSEN queries, including complete-work diagnostics stored by the original procedure. The students themselves sample uniformly from the allowed force-shifted candidates. We preserve the original maximum preparation budget, but actual valid-anchor counts and query counts can differ by source. Four new student fits of 1,000 updates produce the two paired Gaussian models. The harmonic physical model and the two unadapted models reuse their previously audited checkpoints and outputs.

The Gaussian physical models generate 1,536 unoptimized outputs on the existing 24-composition set at 128 network calls per attempt. eSEN evaluation adds 3,072 queries, and GFN2 adds 1,632 attempts including reference inversion checks. One GFN2 calculation fails on a graph-invalid output; every graph-valid output and reference calculation succeeds. The experiment’s audit reconstructs

Table 30: Self-conditioning ablation on 24 compositions. Each run has 768 raw attempts; the valid-output column lists runs 1 and 2. The self-conditioned model outperforms both single-pass training budgets.

Model	Updates	Training passes	Valid outputs	Rate (%)
Single pass	3,000	3,000	363;336	45.51
Single pass	6,000	6,000	277;342	40.30
Self-conditioned	3,000	6,000	435;375	52.73

target proposals and graph restrictions, replays uniform candidate draws and parameter arithmetic, and reparses the structural and physical scores. This is a fixed follow-up on a previously evaluated set, not a new held-out composition test.

For Gaussian, physical adaptation changes energy on paired, jointly valid outputs by -0.01596 eV/atom under eSEN (95% interval $[-0.02182, -0.01021]$) and -0.01465 under GFN2 ($[-0.02041, -0.00929]$). The corresponding force-RMS changes are -0.3795 eV/Å ($[-0.4618, -0.3045]$) and -0.4618 ($[-0.6195, -0.3370]$). The paired physical update therefore also benefits the Gaussian-source generator.

At a force-RMS threshold of 5 eV/Å, GFN2 joint-yield improvements in runs 1 and 2 are 4.56 and 5.34 percentage points for Gaussian and 3.13 and 2.99 for harmonic. After physical adaptation, harmonic minus Gaussian is 16.28 points in run 1 and -2.34 points in run 2, despite a positive pooled difference. Thus the results show both separate source/update benefits and variability between runs. They do not establish a positive statistical interaction between the source and physical update.

X CURVATURE-AWARE LOCAL WORK SAMPLING

The force escort translates every local candidate by the same vector. It does not change the shape of the proposal cloud. Molecular energy can vary more sharply in some directions, such as bond stretching, than in softer collective motions. We test whether estimating this local curvature produces better importance weights at a fixed candidate-query budget, without changing the target distribution.

Fitting local curvature. Flatten the anchor into $3N$ coordinates. Let the columns of $Q_N \in \mathbb{R}^{3N \times d}$ form an orthonormal basis for the centered space, with $d = 3(N - 1)$. A displacement $u \in \mathbb{R}^d$ is expressed in angstroms in this basis, and $Q_N u$ gives the corresponding atom-coordinate displacement. Let f_A be the anchor force in the same basis.

We draw eight independent pilot candidates. Their displacements form columns of D , and the negative changes in force form columns of G . We approximate $G \approx B_A D$ by fitting a symmetric matrix:

$$\|BD - G\|_F^2 + \lambda \|B - \kappa I\|_F^2, \quad \kappa = \max\left(0, \frac{\langle D, G \rangle_F}{\|D\|_F^2}\right), \quad \lambda = 0.01 \|D\|_F^2 / K. \quad (23)$$

Here K counts pilot columns that satisfy the graph restriction. If no pilot survives, we set $B_A = 0$ and recover the original force translation. For a nonempty pilot we floor $\|D\|_F^2$ at 10^{-20} in the scalar and ridge calculations; exactly zero displacements also give zero fitted curvature. The scalar κ estimates an average nonnegative stiffness. The second term shrinks poorly determined directions toward this isotropic estimate. We replace negative eigenvalues of the fitted matrix by zero to obtain a positive semidefinite B_A . The isotropic control uses $B_A = \kappa I$ directly. Pilot information and the fitted map are fixed before drawing production candidates.

Changing the proposal mean and covariance. The fitted quadratic energy is $E_+(A + Q_N u) - E_+(A) \approx -f_A^\top u + u^\top B_A u / 2$. Multiplying its Boltzmann factor by the original Gaussian reference gives another Gaussian with a shifted mean and adjusted covariance. Define $P_A = I + \sigma_A^2 B_A / (k_B T_{\text{phys}})$, a dimensionless positive-definite precision matrix. We map the source displacement u to

$$v = P_A^{-1/2} u + b_A, \quad b_A = P_A^{-1} \frac{\sigma_A^2 f_A}{k_B T_{\text{phys}}}, \quad \log J_A = -\frac{1}{2} \log \det P_A. \quad (24)$$

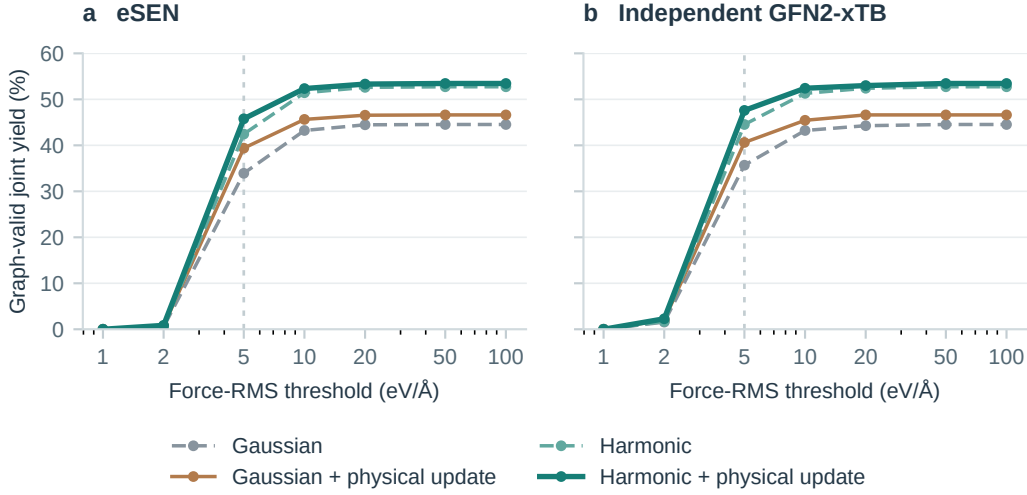


Figure 13: Joint graph-validity and residual-force yield for both sources, with and without the paired physical update. The denominator includes every raw generation attempt. Each threshold is applied to the RMS atomic-force magnitude in $\text{eV}/\text{\AA}$, and both potentials use the same threshold grid. No coordinates are optimized before evaluation.

The factor $P_A^{-1/2}$ contracts perturbations more in stiff directions, while b_A adjusts the force-guided mean shift for curvature. The Jacobian J_A measures volume change in the d independent coordinates. We keep the original reference width and temperature, so only the proposal changes, not the restrained target. With $X = A + Q_N u$ and $Y = A + Q_N v$, complete work is

$$W_A = E_+(Y) - E_+(A) + k_B T_{\text{phys}} [\log q_A(X) - \log q_A(Y) - \log J_A]. \quad (25)$$

Candidates outside the original same-graph support receive zero weight. This is a local use of targeted and escorted work reweighting (Jarzynski, 2002; Vaikuntanathan & Jarzynski, 2008).

Why curvature can improve the weights. Write the actual energy as the fitted quadratic plus a remainder $R_A(v)$. Substitution gives $W_A = -k_B T_{\text{phys}} \log Z_{\text{quad}} + R_A(v)$, where Z_{quad} is the integral of the Gaussian reference times the quadratic Boltzmann factor. The fitted quadratic cancels from the varying part of the work; only the approximation error changes across allowed candidates. If energy is exactly that quadratic, all allowed candidates have equal weights. Tests verify this constant-work identity for an anisotropic quadratic energy and check rotation equivariance of the fitted map.

Pilot comparison at equal query budgets. For each of eight existing training compositions in each run, we take the first four eligible anchors without ranking by energy or ESS. The translation baseline uses 24 fresh production candidates per anchor. Each curvature-fitted map uses at most eight pilot energy/force queries and 16 fresh production candidates. The maximum particle-query budget is therefore the same. Common anchor queries and reflection evaluations are counted separately. We evaluate all production energies, including those of graph-invalid candidates, and do not reuse pilot samples as production samples for the fitted maps.

Directional curvature improves ESS over translation on 61 of 64 anchors. The mean increase is 4.66 effective samples, with composition-bootstrap 95% interval $[3.81, 5.46]$. Same-graph retention is 99.51%, compared with 99.54% for translation. Independent multivariate-Gaussian density calculations reproduce the stored complete-work weights. We then construct full teacher pools for all 349 original eligible anchors using fresh production draws. Translation versus curvature gives mean ESS of 2.32 versus 6.71 in run 1 and 2.33 versus 6.37 in run 2. These results concern the local teacher; the next experiment tests whether they improve the trained generator.

Table 31: Local ESS with complete-work weights at a maximum of 24 particle queries per anchor. Each run uses 32 anchors. Curvature methods reserve eight queries for fitting and use 16 production candidates; translation uses all 24 for production.

Escort	Production particles	ESS, run 1	ESS, run 2
Force translation	24	2.22	1.86
Isotropic curvature	16	3.55	3.94
Directional curvature	16	6.69	6.70

Does the teacher improvement transfer to generation? We train three physical students: uniform sampling of 24 force-translated candidates, complete-work weighting of those same 24 candidates, and complete-work weighting of 16 curvature-adjusted candidates. Each uses the original 1,000-update schedule and reference/anchor choices. We reuse the matched replay model and add the physical-minus-replay parameter difference with coefficient one. All 349 original anchors remain eligible. The two work-weighted methods have equal maximum particle-query budgets; uniform force targets can omit candidate-energy queries. This follow-up changes the candidate-pool size and curvature fit. The main experiments continue to use the original, preselected eight-candidate force model.

Table 32: Generated outputs after training with the three teachers, evaluated on 24 compositions with 1,536 attempts per method across both runs. Energy changes compare a work-weighted student with the uniform-force student on paired, jointly valid outputs. All methods use 128 network evaluations and no geometry optimization.

Teacher	Graph validity (%)	eSEN change (eV/atom)	GFN2 change (eV/atom)
Uniform force, 24 candidates	54.30	0	0
Complete work, 24 candidates	53.39	+0.00959	+0.00845
Curvature work, 16 candidates	53.32	+0.00793	+0.00783

Curvature work minus translation work changes energy by -0.00184 eV/atom under eSEN (95% interval $[-0.00654, 0.00289]$) and -0.00068 under GFN2 ($[-0.00484, 0.00341]$). The two runs have opposite signs. The higher teacher ESS therefore does not establish an additional generation benefit. Uniform force targets give lower energy and force RMS than either work-weighted student under both potentials. Relative to the frozen harmonic generator, this 24-candidate uniform-force model lowers energy by 0.02088 eV/atom under eSEN and 0.01918 under GFN2. The primary method retains the inexpensive force-only recipe.